

US Patent & Trademark Office

Patent Public Search | Text View

United States Patent Application Publication

20250275830

Kind Code

A1

Publication Date

September 04, 2025

Inventor(s)

Abbott; Ryan C. et al.

MODULAR CLAMPS FOR MOUNTING SURGICAL MANIPULATORS

Abstract

A clamping apparatus comprises a body having a mounting location configured to support the surgical manipulator; an articulatable bracket coupled to the body; a first jaw coupled to the body and configured to engage a first portion of the mounting structure; and a second jaw coupled to the bracket and configured to engage a second portion of the mounting structure. The articulatable bracket is configured to articulate relative to the body along a segmented path comprising multiple directions to move the second jaw from an unclamping position to a range of clamping positions. The second jaw is further from the body in the clamping positions than in the unclamping position, and is closer to the first jaw in the clamping positions than in the unclamping position.

Inventors: Abbott; Ryan C. (San Jose, CA), Gomez; Daniel H. (Los Gatos, CA), Steger; John Ryan (Los Gatos, CA)

Applicant: INTUITIVE SURGICAL OPERATIONS, INC. (Sunnyvale, CA)

Family ID: 80115605

Assignee: INTUITIVE SURGICAL OPERATIONS, INC. (Sunnyvale, CA)

Appl. No.: 19/086808

Filed: March 21, 2025

Related U.S. Application Data

parent US continuation 17444158 20210730 parent-grant-document US 12285301 child US 19086808

us-provisional-application US 63063181 20200807

Publication Classification

Int. Cl.: A61B90/57 (20160101); A61B34/00 (20160101); A61B90/35 (20160101)

U.S. Cl.:

**CPC A61B90/57 (20160201); A61B34/70 (20160201); A61B90/35 (20160201);
A61B2090/571 (20160201)**

Background/Summary

CROSS REFERENCE TO RELATED APPLICATION [0001] This application is a continuation application of application Ser. No. 17/444,158, filed Jul. 30, 2021, which claims the benefit of U.S. Provisional Application No. 63/063,181, filed Aug. 7, 2020 (now expired), each of which is incorporated by reference herein in their entirety.

TECHNICAL FIELD

[0002] Aspects of the present disclosure generally relate to devices and methods for performing a computer-assisted teleoperated procedure and, more specifically, to devices for arranging and mounting various tools for the computer-assisted teleoperated procedure.

BACKGROUND

[0003] Minimally invasive medical techniques are intended to reduce an amount of tissue that is damaged during medical procedures, thereby reducing patient recovery time, discomfort, and harmful side effects. Such minimally invasive techniques may be performed through natural orifices in a patient anatomy or through one or more surgical incisions. An operator (e.g., a physician) may insert minimally invasive medical instruments (surgical, diagnostic, therapeutic, biopsy instruments, etc.) through these natural orifices or incisions to reach a target tissue location. Robotic medical systems allow a user to control such medical instruments via a manipulator to which the instrument is mounted. One such minimally invasive technique is to mount a modular manipulator assembly to a surgical table with a clamp and independently move one or more instruments to a region of interest within the patient anatomy to perform a medical procedure. Control of such a manipulator assembly by an operator involves the management of several degrees of freedom including at least the management of insertion, retraction, and roll of the instrument with respect to the patient anatomy, as well as articulation of an instrument end effector.

[0004] Modular systems can use arrangeable clamps to mount various tools to a mounting structure. Manufacturers of systems having modular equipment mounting structures (e.g., operating tables, side carts, etc.) may specify their mounting structures with differing dimensions, which requires the arrangeable clamps to have corresponding features to accommodate the particular configuration of the mounting structure. In practice, a procurement order for a modular system using arrangeable clamps must specify the dimensions of the mounting structure (e.g., the height and width of a mounting rail), and consequently, the modular system is not usable with other mounting structures having dimensions the arrangeable clamps cannot accommodate.

SUMMARY

[0005] In accordance with embodiments of the present disclosure, a clamping apparatus is provided to support a robotic manipulator and couple to mounting structures of different sizes (e.g., structures having different heights, widths, and/or depths). The clamping apparatus has one or more jaws configured to move along a nonlinear path to accommodate different rail sizes.

[0006] In accordance with an embodiment of the present disclosure, a system is provided. The system generally includes a surgical manipulator configured to control motion of a surgical tool, and a clamping apparatus having a mounting location configured to support the surgical manipulator and couple to a mounting structure. The clamping apparatus may include a body, a first

jaw coupled to the body and configured to engage a first portion of the mounting structure, and a second jaw coupled to the body and configured to engage a second portion of the mounting structure. At least one of the first jaw and the second jaw may be movably coupled to the body and configured to move along a non-linear path to clamp mounting structures of different sizes.

[0007] In accordance with an embodiment of the present disclosure, a clamping apparatus is provided. The clamping apparatus generally includes a body having a first cam slot having a non-linear path and a second cam slot, a first jaw fixed to the body and configured to engage a portion of a mounting structure, and a second jaw movably coupled to the body within the first and second cam slots. The clamping apparatus may further include a lever pivotably coupled to the body and the second jaw and movable between a first position, wherein the clamping apparatus has an unclamped mechanical state, to a second position, wherein the clamping apparatus has a clamped mechanical state, wherein articulation of the lever away from the first position toward the second position may cause the second jaw to move within the first and second cam slots toward the first jaw, and wherein the non-linear path of the first cam slot may initially cause the second jaw to move outwardly away from the body as the second jaw moves toward the first jaw. The clamping apparatus may further include a biasing member positioned between the body and the lever and configured to bias the second jaw against the mounting structure in the first position, wherein articulation of the lever toward the second position after the second jaw contacts the mounting structure may cause the biasing member to compress along a compliance direction.

[0008] In accordance with an embodiment of the present disclosure, a clamping apparatus is provided. The clamping apparatus generally includes a body, a first jaw fixed to the body and configured to engage a portion of a mounting structure, an articulable bracket movably coupled to the body, and a second jaw movably coupled to the articulable bracket. The clamping apparatus may further include a lever pivotably coupled to the body and the second jaw and movable between a first position, wherein the clamping apparatus is unclamped, to a second position, wherein the clamping apparatus is clamped, wherein articulation of the lever away from the first position toward the second position may cause the second jaw to move toward the first jaw. The clamping apparatus may further include a first biasing member positioned between the body and the lever and configured to bias the lever toward the first position, wherein articulation of the lever toward the second position after the second jaw contacts the mounting structure may cause the first biasing member to compress along a first compliance direction. The clamping apparatus may further include a second biasing member positioned between the articulable bracket and the second jaw and configured to bias the second jaw away from the articulable bracket, wherein articulation of the lever toward the second position after the second jaw contacts the mounting structure may cause the second biasing member to compress along a second compliance direction.

[0009] In accordance with an embodiment of the present disclosure, a clamping apparatus is provided. The clamping apparatus generally includes a first jaw fixed to a body and configured to engage a portion of a mounting structure, a second jaw movably coupled to the body and having an angled surface to accommodate a range of widths of the mounting structure, and a motor operably coupled to the body and the second jaw to move the second jaw between a first position, wherein the clamping apparatus has an unclamped mechanical state, and a second position, wherein the clamping apparatus has a clamped mechanical state, wherein actuation of the motor in the first position may cause the second jaw to move toward the first jaw. The clamping apparatus may further include a sensor configured to detect the mounting structure, wherein the motor may be operable when the sensor detects the mounting structure.

[0010] In accordance with an embodiment of the present disclosure, a clamping apparatus is provided. The clamping apparatus generally includes a body, a first jaw slidably coupled to the body and having a first surface configured to engage an upper portion of a mounting structure and a second surface configured to engage a lateral portion of the mounting structure, a second jaw movably coupled to the body and having a third surface configured to engage a lower portion of the

mounting structure and a fourth surface configured to engage the lateral portion of the mounting structure engaged by the first surface of the first jaw, and a motor operably coupled to the body and the second jaw to move the second jaw between a first position, wherein the clamping apparatus is unclamped, and a second position, wherein the clamping apparatus is clamped, wherein actuation of the motor in the first position may cause the second jaw to move toward the first jaw. The clamping apparatus may further include a sensor configured to detect the mounting structure, wherein the motor may be operable when the sensor detects the mounting structure.

[0011] In accordance with any of the embodiments disclosed herein, the movement of the second jaw may be arcuate.

[0012] In accordance with any of the embodiments disclosed herein, the mounting structure may comprise a rail, and wherein the first jaw may be configured to abut an upper portion of the rail and the second jaw may be configured to abut a lower portion of the rail when the clamping apparatus is clamped to the rail.

[0013] In accordance with any of the embodiments disclosed herein, the non-linear path of the first cam slot may be configured to move the second jaw inwardly toward the rail as the second jaw moves toward the first jaw after the first cam slot initially causes the second jaw to move outwardly away from the body.

[0014] In accordance with any of the embodiments disclosed herein, the movement of the second jaw within the first and second cam slots may be locked when the lever is in the second position.

[0015] In accordance with any of the embodiments disclosed herein, the movement of the second jaw within the first and second cam slots may be locked by a toothed interface between the lever and the body, fixing the biasing member in the compliance direction.

[0016] In accordance with any of the embodiments disclosed herein, the biasing member may be a first biasing member and the compliance direction is a first compliance direction, the clamping apparatus may further comprise an articulable bracket positioned between the second jaw and the body and configured to allow movement of the second jaw in a second compliance direction independent of the movement within the first and second cam slots, and the movement of the second jaw with respect to the articulable bracket may be biased by a second biasing member.

[0017] In accordance with any of the embodiments disclosed herein, the movement of the second jaw with respect to the articulable bracket may be locked when the lever is in the second position.

[0018] In accordance with any of the embodiments disclosed herein, the movement of the second jaw with respect to the articulable bracket may be locked by a toothed arm interfacing with the second jaw, fixing the second biasing member in the second compliance direction.

[0019] In accordance with any of the embodiments disclosed herein, the first jaw may have an angled surface to accommodate a range of widths of the mounting structure.

[0020] In accordance with any of the embodiments disclosed herein, the lever may be retained in the second position by an over-center configuration.

[0021] In accordance with any of the embodiments disclosed herein, the lever may have an articulable portion that interfaces with the body to retain the lever in the second position until the articulable portion is rotated.

[0022] In accordance with any of the embodiments disclosed herein, one or more of the first jaw and second jaw may have rounded edges to reduce abrasion of a drape positioned between the clamping apparatus and the mounting structure.

[0023] In accordance with any of the embodiments disclosed herein, the body may further comprise a mounting feature configured to receive a component and releasably couple the component to the clamping apparatus.

[0024] In accordance with any of the embodiments disclosed herein, the body may further comprise a cable receptacle configured to receive a cable therein.

[0025] In accordance with any of the embodiments disclosed herein, the body may further comprise a connector cavity at an end of the cable receptacle to receive a connector therein.

[0026] In accordance with any of the embodiments disclosed herein, the apparatus may further comprise a sensor configured to detect load on the clamping apparatus to signal an improper connection.

[0027] In accordance with any of the embodiments disclosed herein, the apparatus may further comprise a lamp configured to illuminate when the clamping apparatus is properly clamped to the mounting structure

[0028] In accordance with any of the embodiments disclosed herein, the sensor may be a first sensor, and wherein the clamping apparatus may further comprise a second sensor configured to stop actuation of the motor when the second and fourth surfaces contact the lateral portion of the mounting structure.

[0029] In accordance with any of the embodiments disclosed herein, the apparatus may further comprise an articulable bracket positioned between the second jaw and the body, wherein the second jaw may be slidably coupled to the articulable bracket such that the second jaw is independently movable with respect to the articulable bracket as the articulable bracket is movable with respect to the body

[0030] In accordance with any of the embodiments disclosed herein, the articulable bracket may cause the second jaw to travel along an arcuate path toward the first jaw.

[0031] In accordance with any of the embodiments disclosed herein, the second jaw may be slidable with respect to the articulable bracket along a linear path disposed at an angle from the lateral portion of the mounting structure such that as the motor actuates the second jaw toward the second position, the fourth surface of the second jaw moves toward the lateral portion of the mounting structure

[0032] In accordance with any of the embodiments disclosed herein, the first jaw may be slidable with respect to the body along a linear path disposed at an angle from the lateral portion of the mounting structure such that as the motor actuates the second jaw toward the second position, the second surface of the first jaw moves toward the lateral portion of the mounting structure.

Description

DESCRIPTION OF THE DRAWINGS

[0033] Many aspects of the present disclosure can be better understood with reference to the detailed description along with the following drawings. The components in the drawings are not necessarily drawn to scale. Instead, emphasis is placed on illustrating clearly the principles of the present disclosure. Furthermore, components can be shown as transparent in certain views for clarity of illustration only and not to indicate that the component is necessarily transparent. Components may also be shown schematically.

[0034] FIG. 1A is a simplified diagram of a medical system configured in accordance with an embodiment of the present disclosure.

[0035] FIG. 1B is a perspective view of a structural representation of the medical system of FIG. 1A.

[0036] FIGS. 2A and 2B are front-right-top and rear-right-top perspective views, respectively, of a modular clamp assembly configured in accordance with embodiments of the present disclosure.

[0037] FIGS. 2C-2F are cross-sectional right side views of the modular clamp assembly of FIGS. 2A and 2B, with the cross-sections taken along corresponding positions shown in FIGS. 2A and 2B.

[0038] FIG. 2G shows a detail view of certain components of the modular clamp assembly of FIGS. 2A and 2B with the body and the manual release lever hidden for clarity.

[0039] FIGS. 3A and 3B are front-right-top and rear-right-top perspective views, respectively, of a modular clamp assembly configured in accordance with embodiments of the present disclosure.

[0040] FIGS. 3C-3F are cross-sectional right side views of the modular clamp assembly of FIGS. 3A and 3B, with the cross-sections taken along corresponding positions shown in FIGS. 3A and 3B.

[0041] FIGS. 4A and 4B are front-right-top and rear-right-top perspective views, respectively, of a modular clamp assembly configured in accordance with embodiments of the present disclosure.

[0042] FIGS. 4C-4H are cross-sectional right side views of the modular clamp assembly of FIGS. 4A and 4B, with the cross-sections taken along corresponding positions shown in FIGS. 4A and 4B.

[0043] FIGS. 5A and 5B are front-right-top and rear-right-top perspective views, respectively, of a modular clamp assembly configured in accordance with embodiments of the present disclosure.

[0044] FIGS. 5C-5H are cross-sectional right side views of the modular clamp assembly of FIGS. 5A and 5B, with the cross-sections taken along corresponding positions shown in FIGS. 5A and 5B.

[0045] In the specification, it should be appreciated that like reference numerals are used to identify like elements illustrated in one or more of the figures for purposes of illustrating embodiments of the present disclosure and not for purposes of limiting the same.

DETAILED DESCRIPTION

[0046] The present disclosure generally relates to a modular clamp for mounting a tool (e.g., a surgical arm) to a structure. The modular clamps disclosed herein may have adapter features that are configured to allow mounting of various tools to the modular clamp, either rigidly or with one or more degrees of freedom (DOF). The modular clamps may be configured to accommodate different mounting structures, e.g., rectangular mounting rails having a range of heights, widths, and/or depths. In addition, modular clamps configured in accordance with the present disclosure may be adapted to accommodate further shapes and sizes of mounting structures, such as rails having different sizes. For example, the clamps may be used to accommodate different rails with different rail sizes. As another example, the clamps may be used to accommodate different portions of a single rail where the different portions have different rail sizes. The clamps may also allow repositioning of one or more components attached to the clamp without removal of the component from the clamp, such as by one or more accessory mounting locations, among other features. Such mounting locations may have coupling features, such as detents or other retention features, to retain the other components in the mounting location until removal from the clamp is desired, e.g., manual rotation of the component to release, overcoming the detent, etc. Mounting location(s) may be illuminated to aid in coupling the component to the clamp. The clamps may include various openings, slots, and indices for one or more auxiliary features, such as cable routing, connector mounting, positioning of the clamp, etc. In some embodiments with cable routing, the mounting location may include a connector to connect a signal cable routed through the clamp to a signal cable of the mounted component upon coupling. The clamps having electrical clamping features may also include a manual release lever that can be pivoted to release the clamp from the mounting structure.

[0047] Certain mounting systems described throughout this disclosure are appropriately configured to retrofit a conventional operating table in a manner that provides a robust and stable mounting point for securing a teleoperated surgical arm to the table. Continued development of teleoperated surgical systems has led to the implementation of individual surgical arms that can be received and supported by a relatively compact interface. The interface can be a part of a portable module positionable near a patient's body and secured to an operating room structure (e.g., an operating table) by medical staff prior to a surgical procedure. This modular configuration provides an alternative to other arrangements where one or more surgical arms are attached to a comparatively large support assembly on a stand-alone structure. Accordingly, one advantage that may be gained from implementing such a modular system is more efficient space utilization in an operating room.

[0048] Various embodiments described within this disclosure are derived from a realization that

many existing operating tables were not designed to support a teleoperated surgical arm. Nevertheless, it may be desirable and advantageous to use the operating table's side rails for mounting a teleoperated robotic surgical arm and/or other associated tools for multiple reasons. First, the rails are conveniently located on a side of the operating table, and therefore proximate the target location of the surgical arm, i.e., near the patient. Second, the side rails allow a variety of objects to be readily attached to the operating table by clamps (or other similar coupling devices). [0049] Accordingly, the present disclosure describes various mounting systems that can be attached to an operating table in a manner that adds stability to the standard side rails, offering a robust and sturdy mounting point to secure a surgical arm or associated tool during a surgical procedure. Employing the mounting systems described herein enables the use of surgical arms with pre-existing operating tables, which allows a medical facility to obtain the benefits of teleoperated minimally invasive surgery at lower cost.

[0050] Embodiments of the modular clamp assemblies described below include various clamping and releasing arrangements with jaws having various degrees of freedom. In this regard, the embodiments can include automated electro-mechanical actuation, manual electro-mechanical actuation, and/or manual mechanical actuation. Although some of the embodiments are shown with or without electro-mechanical components, any of the embodiments described herein are adaptable for either electro-mechanical or mechanical actuation. Generally described, the modular clamp assemblies described below have the following configurations: (A) the arrangement shown in FIGS. 2A-2G is electro-mechanical having an upper jaw with one degree of freedom and a lower jaw with two degrees of freedom; (B) the arrangement shown in FIGS. 3A-3F is electro-mechanical having a lower jaw with one degree of freedom; (C) the arrangement shown in FIGS. 4A-4H is mechanical having a lower jaw with two degrees of freedom; and (D) the arrangement shown in FIGS. 5A-5H is mechanical having a lower jaw with one degree of freedom. In other embodiments of the present disclosure, each of the arrangements disclosed herein may be adapted such that the upper jaw and/or lower jaw have any number of degrees of freedom.

[0051] The present disclosure describes various instruments and portions of instruments in terms of their state in three-dimensional space. As used herein, the term position refers to the location of an object or a portion of an object in a three-dimensional space (e.g., three degrees of translational freedom along Cartesian X-, Y-, and Z-coordinates). As used herein, the term orientation refers to the rotational placement of an object or a portion of an object (e.g., three degrees of rotational freedom, such as roll, pitch, and yaw). As used herein, the term pose refers to the position of an object or a portion of an object in at least one degree of translational freedom and to the orientation of that object or portion of the object in at least one degree of rotational freedom (e.g., up to six total degrees of freedom). As used herein, the term shape refers to a set of poses, positions, or orientations measured along an object.

[0052] FIG. 1A is a simplified diagram of a medical system **100** ("system **100**"). In some embodiments, the system **100** may be suitable for use in, for example, surgical, teleoperated surgical, diagnostic, therapeutic, or biopsy procedures. While some embodiments are provided herein with respect to such procedures, any reference to medical or surgical instruments and medical or surgical methods is intended as non-limiting. The systems, instruments, and methods described herein may be used for animals, human cadavers, animal cadavers, portions of human or animal anatomy, non-surgical diagnosis, as well as for industrial systems and general robotic, general teleoperational, or robotic medical systems. For example, the clamps, systems, and methods described herein may be used for non-medical purposes including industrial uses, general robotic uses, and sensing or manipulating non-tissue work pieces. Other example applications involve cosmetic improvements, imaging of human or animal anatomy, gathering data from human or animal anatomy, and training medical or non-medical personnel. Additional example applications include use for procedures on tissue removed from human or animal anatomies (without return to a human or animal anatomy), and performing procedures on human or animal

cadavers. Further, these techniques can also be used for surgical and nonsurgical medical treatment or diagnosis procedures.

[0053] As shown in FIG. 1A, the system **100** generally includes a plurality of manipulator assemblies **102**. Although three manipulator assemblies **102** are illustrated in the embodiment of FIG. 1A, in other embodiments, more or fewer manipulator assemblies may be used. The exact number of manipulator assemblies will depend on the medical procedure and the space constraints within the operating room, among other factors. Multiple user control systems **106** may be co-located or they may be positioned in separate locations. Multiple user control systems **106** allow more than one operator to control one or more teleoperated manipulator assemblies in various combinations.

[0054] The manipulator assembly **102** is used to operate a medical instrument **104** (e.g., a surgical instrument or an image capturing device) in performing various procedures on a patient P. The medical instrument **104** may be sterile prior to being used in the various procedures. The manipulator assembly **102** may be teleoperated, non-teleoperated, or a hybrid teleoperated and non-teleoperated assembly with select degrees of freedom of motion that may be motorized and/or teleoperated, and select degrees of freedom of motion that may be non-motorized and/or non-teleoperated. In some embodiments, the manipulator assembly **102** may be mounted near an operating or surgical table T, or the manipulator assembly **102** may be mounted directly to the table T or to a rail coupled to the table T. In various other embodiments, the manipulator assembly **102** may be mounted to a manipulating system (e.g., a patient-side cart). The manipulating system may be separate from and spaced from the table T in the operating room and may be independently movable relative to the table T.

[0055] The manipulator assembly **102** may be mounted to a ceiling, floor, and/or wall of the operating room. In embodiments in which a plurality of manipulator assemblies **102** are employed, one or more of the manipulator assemblies **102** may support surgical instruments, and another of the manipulator assemblies may support an image capturing device such as a monoscopic or stereoscopic endoscope. In such embodiments, one or more of the manipulator assemblies **102** may be mounted to any structure or in any manner as described above. For example, one manipulator assembly **102** may be mounted to the table T and another manipulator assembly **102** may be mounted to a manipulating system.

[0056] A user control system **106** allows an operator (e.g., a surgeon or other clinician, as illustrated in FIG. 1A) to view the interventional site and to control the manipulator assembly **102**. In some examples, the user control system **106** is a surgeon console, which can be located in the same room as the operating or surgical table T, such as at the side of a table on which the patient P is located. However, the operator O can be located in a different room or a completely different building from patient P. The user control system **106** generally includes one or more input devices for controlling the manipulator assembly **102**. The input devices may include any number of a variety of devices, such as joysticks, trackballs, data gloves, trigger-guns, hand-operated controllers, voice recognition devices, body motion or presence sensors, and/or the like. The input devices may be provided with the same degrees of freedom as the associated medical instrument **104** to provide the operator O a strong sense of directly controlling the medical instrument **104**. In this regard, the input devices may provide the operator O with telepresence: the perception that the input devices are integral with medical instrument **104**.

[0057] The input devices may have more or fewer degrees of freedom than the associated medical instrument **104** and still provide the operator O with telepresence. The input devices may optionally be manual input devices that move with six degrees of freedom, and may also include an actuatable handle for actuating instruments (for example, for closing grasping jaws, applying an electrical potential to an electrode, delivering a medicinal treatment, etc.).

[0058] The manipulator assembly **102** may support the medical instrument **104** and may include a kinematic structure of one or more non-servo controlled links (e.g., one or more links that are

manually positioned and locked in place), and/or one or more servo controlled links (e.g., one or more links that are controlled in response to commands from a control system), and a manipulator. The manipulator assembly **102** may optionally include a plurality of actuators or motors (described in greater detail below) that drive inputs on the medical instrument **104** in response to commands from the control system (e.g., a control system **110**). The actuators may optionally include drive systems that when coupled to the medical instrument **104** may advance the medical instrument **104** into a naturally or surgically created anatomic orifice.

[0059] Other drive systems may move the distal end of the medical instrument **104** in multiple degrees of freedom, which can include three degrees of linear motion (e.g., linear motion along the X, Y, Z Cartesian axes), and three degrees of rotational motion (e.g., rotation about the X, Y, Z Cartesian axes). Additionally, the actuators can be used to actuate an articulable end effector of the medical instrument **104**, e.g., for grasping tissue in the jaws of a biopsy device. Actuator position sensors such as resolvers, encoders, potentiometers, and other mechanisms may provide sensor data to the system **100** describing the rotation and orientation of the shafts of the actuator. Such position sensor data may be used to determine motion of the objects manipulated by the actuators. The manipulator assembly **102** may position its held instrument such that a pivot point occurs at the entry aperture into the patient. The manipulator assembly **102** may then manipulate its held instrument so that the instrument may be pivoted about the pivot point, inserted into and retracted out of the entry aperture, and/or rotated about its shaft axis.

[0060] The system **100** may also include a display system **108** for displaying an image or representation of the surgical site and the medical instrument **104**. The display system **108** and the user control system **106** may be oriented so the operator O can control the medical instrument **104** and the user control system **106** with the perception of telepresence. The medical instrument **104** may include a visualization system, which may include a viewing scope assembly that records a concurrent or real-time image of a surgical site and provides the image to the operator O and/or other operators or personnel through one or more displays of the system **100**, such as one or more displays of the display system **108**. The concurrent image may be, for example, a two- or three-dimensional image captured by an endoscope positioned within the surgical site. The visualization system may be implemented as hardware, firmware, software, or a combination thereof that interact with or are otherwise executed by one or more computer processors that may include the processors of the control system **110**. The display system **108** may present images of a surgical site recorded pre-operatively or intra-operatively using image data from imaging technology such as, computed tomography (CT), magnetic resonance imaging (MRI), fluoroscopy, thermography, ultrasound, optical coherence tomography (OCT), thermal imaging, impedance imaging, laser imaging, nanotube X-ray imaging, and/or the like. The pre-operative or intra-operative image data may be presented as two-dimensional, three-dimensional, or four-dimensional (including, e.g., time-based or velocity-based information) images, and/or as images from models created from the pre-operative or intra-operative image data sets.

[0061] The system **100** may also include the control system **110**. The control system **110** may include at least one memory (not shown) and at least one computer processor (not shown) for effecting control between the medical instrument **104**, the user control system **106**, and the display system **108**. The control system **110** also includes programmed instructions (e.g., a non-transitory machine-readable medium storing the instructions) to implement some or all the methods described in accordance with aspects of the present disclosure disclosed herein, including instructions for providing information to the display system **108**. While the control system **110** is shown as a single block in the simplified schematic of FIG. 1A, the system may include two or more data processing circuits with one portion of the processing optionally being performed on or adjacent to the manipulator assembly **102**, another portion of the processing being performed at the user control system **106**, etc. The processors of the control system **110** may execute instructions comprising instruction corresponding to processes disclosed herein and described in more detail below. Any of

a wide variety of centralized or distributed data processing architectures may be employed. Similarly, the programmed instructions may be implemented as a number of separate programs or subroutines, or they may be integrated into a number of other aspects of the robotic medical systems described herein. In one embodiment, the control system **110** supports wireless communication protocols such as Bluetooth, IrDA, HomeRF, IEEE 802.11, DECT, and Wireless Telemetry.

[0062] In some embodiments, a communication is sent from the control system **110** to the manipulator assembly **102**. Additionally, status information regarding testing of the manipulator assembly **102** may be sent from a sterile storage container (not shown) to the control system **110**.

[0063] This status information is used to optimize the performance of the system **100** by indicating an operational status of one or more components of the manipulator assembly **102**. The status information may additionally be received by the operator O, a surgeon, and/or any other suitable personnel. The status information may also be received by a hospital information system, a patient information portal, a surgical information database, and/or any other suitable information system or database. In some embodiments, the status information is sent to a manufacturer of the manipulator assembly **102** to indicate whether the manipulator assembly **102** or any other component of the system requires maintenance.

[0064] Movement of the manipulator assembly **102** may be controlled by the control system **110** such that a shaft or intermediate portion of instruments mounted to the manipulator assemblies **102** are constrained to safe motions through minimally invasive surgical access sites or other apertures. Such motion may include, for example, axial insertion of a shaft through an aperture site, rotation of the shaft about its axis, and pivotal motion of the shaft about a pivot point adjacent the access site. In some cases, the risk of excessive lateral motion of the shaft that might potentially cause hazardous forces on the tissues adjacent the aperture or enlarge the access site inadvertently is inhibited. Some or all of such constraint on the motions of the manipulator assemblies **102** at the access sites may be imposed using mechanical manipulator joint linkages that inhibit improper motions, or may in part or in full be imposed using data processing and control techniques. In some embodiments, the control system **110** may receive force and/or torque feedback from the medical instrument **104**. Responsive to the feedback, the control system **110** may transmit signals to the user control system **106**. In some examples, the control system **110** may transmit signals instructing one or more actuators of the manipulator assembly **102** to move the medical instrument **104**.

[0065] FIG. **1B** is a perspective view of a patient coordinate space **120** including first and second teleoperated surgical manipulator assemblies **130** and **150** mounted on a side of a surgical table T according to one or more aspects of the present disclosure. The first and second manipulator assemblies **130** and **150** may be used as the manipulator assembly **102** (FIG. **1A**) in a medical procedure performed with the system **100** and controlled by the control system **110**. The first and second manipulator assemblies **130** and **150** may be used in procedures involving traditional manually operated minimally invasive surgical instruments, such as manual laparoscopy. While only the first and second manipulator assemblies **130** and **150** are depicted, more than two (e.g., three, four, five, six, and more than six) or fewer than two manipulator assemblies can be included in some configurations of the system **100**.

[0066] A mounting structure **122** (e.g., an equipment rail, an equipment channel, etc.) may be configured such that the first and second manipulator assemblies **130** and **150** can be mounted to the table T during the medical procedure. The mounting structure **122** may differ in size and/or shape (e.g., height, width, etc.) between manufacturers, models, vintages, etc. In some embodiments, the mounting structure **122** of a single table T may have portions with different sizes and/or shapes. For example, the mounting structure **122** may have a first portion with a first size and/or shape and a second portion with a second size and/or shape different than the first portion, as well as additional other portions with different sizes and/or shapes. As described in more detail below, components mounting to the mounting structure **122**, such as the embodiments described

herein, can be configured to accommodate different sizes and shapes of the mounting structure **122** without changing configurations and/or manual adjustment. As referenced herein, the mounting structure **122** may generally have an aspect ratio where the height of the mounting structure is greater than the width of the mounting structure, e.g., a rail with a rectangular profile as shown in FIGS. 3C, 3D, etc. In this regard, the mounting structures described herein may be rotated 90° from the orientation of the mounting structure shown in FIG. 1B.

[0067] In some embodiments, the mounting structure **122** may be attached to a manipulating system (e.g., a patient-side cart, a side table, etc.). The first manipulator assembly **130** may operably move a first instrument **132** within the patient coordinate space **120**, and the second manipulator assembly **150** may operably move a second instrument **152** within the patient coordinate space **120**. The first and second instruments **132** and **152** can be sterilized prior to use in a medical procedure. In this regard, the sterile boundary may be positioned above, below, or intermediate to the clamps described in the embodiments of the present disclosure. One or more drapes (not shown) may be used to maintain a sterile field, and the edges of the clamps described herein may be configured to reduce sharp edges that could abrade or otherwise tear the drapes. Various mounting arrangements and configurations manipulator assemblies are described in U.S. application Ser. No. 16/408,077 (filed May 9, 2019, titled “Mounting Teleoperated Surgical Arms”), which is incorporated by reference herein in its entirety.

[0068] The first manipulator assembly **130** may include a first manipulator **134**, a first link **138**, and a first drive unit **140**. The second manipulator assembly **150** may include a second manipulator **154**, a second link **158**, and a second drive unit **160**. The first instrument **132** may be coupled to the first drive unit **140**, and the second instrument **152** may be coupled to the second drive unit **160**. In some embodiments, the first drive unit **140** is, for example, a standalone unit including a system of drive mechanisms (not shown, e.g., motors). The first drive unit **140** may be operated to control motion of the first instrument **132** in multiple DOFs when the first instrument **132** is mounted to the first drive unit **140**. The second drive unit **160** may be similarly configured for operation of the second instrument **152**. The first drive unit **140** may be coupled to the first manipulator **134**, and the second drive unit **160** may be coupled to the second manipulator **154**. The first manipulator **134** may be movably coupled to the first link **138**, and the second manipulator **154** may be movably coupled to the second link **158**.

[0069] Any one or more of the components of the first manipulator **134**, the second manipulator **154**, the first drive unit **140**, and/or the second drive unit **160** may be teleoperated. Further, many of the components (the first and second instruments **132** and **152**; the first and second drive units **140** and **160**; the first and second manipulators **134** and **154**; and the first and second links **138** and **158**) may be sterilized prior to use in a medical procedure. Additionally, one or more of the input devices (e.g., joysticks, trackballs, data gloves, trigger-guns, hand-operated controllers, voice recognition devices, body motion or presence sensors, and/or the like), which may be used for controlling the manipulator assembly **102** (FIG. 1A), may be sterilized prior to use in a medical procedure.

[0070] In embodiments where the first manipulator assembly **130** is mounted to the mounting structure **122** of the surgical table T, the manipulator assembly **130** may be coupled with a first coupling member **142** and a first clamp **144** to the surgical table T. The first coupling member **142** may be a joint (e.g., a ball joint, a spherical ball joint, a prismatic joint, a gimbal, etc.). The second manipulator assembly **150** may be coupled to the surgical table T by a second coupling member **162** and a second clamp **164**. A first housing **136** may be coupled to the first coupling member **142**, and a second housing **156** may be coupled to the second coupling member **162**. In some embodiments, the first manipulator assembly **130** includes the first coupling member **142**, the first clamp **144**, and the first housing **136**, and the second manipulator assembly **150** includes the second coupling member **162**, the second clamp **164**, and the second housing **156**.

[0071] The first manipulator assembly **130** may be coupled to the mounting structure **122** of the

surgical table T by the first clamp **144** (which may be a support component and/or a support structure) may kinematically support the first manipulator **134** and, therefore, the first manipulator assembly **130** before, during, and/or after a surgical procedure. The first clamp **144** may translate along the mounting structure **122** to allow the position of the first manipulator assembly **130** to be moved or readjusted relative to the table T and the patient P. The first clamp **144** may be coupled to the table T via the mounting structure **122**, or may be coupled directly to the table T. In other embodiments, the first clamp **144** can be coupled directly to the first housing **136** or the first link **138** without the first coupling member **142**. In some embodiments, the first clamp **144** can be coupled to a separate cart that is positioned near or attached to the table T.

[0072] The second manipulator assembly **150** may be coupled to the mounting structure **122** of the surgical table T by the second clamp **164**. The second clamp **164** (which may be a support component and/or a support structure) may kinematically support the second manipulator **154** and, therefore, the second manipulator assembly **150** before, during, and/or after a surgical procedure. The second clamp **164** may translate along the mounting structure **122** to allow the position of the second manipulator assembly **150** to be moved or readjusted relative to the table T and the patient P. The second clamp **164** may be coupled to the table T via the mounting structure **122**, or may be coupled directly to the table T. In other embodiments, the second clamp **164** can be coupled directly to the second housing **156** or the second link **158** without the second coupling member **162**. In some embodiments, the second clamp **164** can be coupled to a separate cart that is positioned near or attached to the table T.

[0073] Movement of the first and second manipulator assemblies **130** and **150** (e.g., insertion, rotation, translation, etc.) can be actuated by either robotic control or by manual intervention by the operator O. For example, the position and orientation of the first and second manipulator assemblies **130** and **150** may be maintained in a stationary configuration using one or more brakes. However, depression of one or more buttons and/or switches may release one or more corresponding brakes, allowing the operator O to manually reposition either of the first and second manipulator assemblies **130** and **150**, to thereby position the medical instrument **104**. One or more adjustments may also be controlled by one or more actuators (e.g., motors) such that the operator O may use a button or switch to actuate a motor to alter either of the first and second manipulator assemblies **130** and **150** in a desired manner to position either of the first and second manipulator assemblies **130** and **150** in the optimal position and orientation.

[0074] FIGS. 2A and 2B are front-right-top and rear-right-top perspective views, respectively, of a modular clamp assembly (“clamp **200**”) configured in accordance with embodiments of the present disclosure. The clamp **200** is configured to be removably attached to a mounting structure such that various components and tools can be mounted to or supported by the clamp **200** and modularly arranged with respect to the mounting structure. In some embodiments, the mounting structure is a rail of the table T shown in FIGS. 1A and 1B, and the clamp **200** is configured to support one or more of the components described above in reference to the first and second manipulator assemblies **130** and **150** (FIG. 1B). In this regard (and with reference to FIG. 1B) the clamp **200** may be operatively associated with the instruments **132** and **152**, the manipulators **134** and **154**, the housings **136** and **156**, the links **138** and **158**, the drive units **140** and **160**, the coupling members **142** and **162** in the system **100**, and may replace the illustrated clamps **144** and **164** of FIG. 1B. In other embodiments, any rigid, adjustable, and/or positionable system may be mounted to the table T or other manipulating system (e.g., a patient-side cart) using the clamp **200**. As shown, such clamps **200** may allow adaptation of the configuration of one or more manipulator assemblies tailored to the anatomy of the patient P, the procedure to be performed, the preference of the operator O, the layout of the patient coordinate space **120**, etc.

[0075] The clamp **200** generally includes a body **202** configured to at least partially surround and protect various components of the clamp **200**, provide component positioning and dynamic features, and include one or more accessory mounting locations, among other features. The body

202 may include a distal mounting location **204** configured to couple to other components of the system **100**, e.g. the coupling members **142** and **162**. In other embodiments, the body **202** includes any number of mounting locations to adapt the clamp **200** for mounting various other components. Examples of various components that may be coupled to the distal mounting location **202** or other mounting locations on the body **202** are described in U.S. Pub. No. 2019/0223967 (filed Oct. 4, 2017, titled “Computer-Assisted Teleoperated Surgery Systems and Methods”) and which is incorporated by reference herein in its entirety. In any of the embodiments of the present disclosure, the mounting locations have coupling features, such as detents or other retention features, to retain mounted components in the mounting location until removal is desired, e.g., manual rotation of the component to release, overcoming the detent, etc. The clamps described herein may also be removed from the mounting structure (e.g., the mounting structure **122**) without removing the component coupled to the clamp, and the mounting location may be illuminated to aid in coupling the component to the clamp. The body **202** may include other features, such as various openings, slots, and indices (not shown with respect to the clamp **200**, but explained in greater detail below in reference to other embodiments of the present disclosure) for one or more auxiliary features of the clamp **200**, such as cable routing, connector mounting, positioning of the clamp, etc. In some embodiments with cable routing, the mounting location may include a connector to connect a signal cable routed through the clamp to a signal cable of the mounted component upon coupling. The clamp **200** may also include a manual release lever **250** (FIG. 2B) that can be pivoted in the direction of arrow O to release the clamp **200** from the mounting structure. The manual release lever **250** will be explained in greater detail with reference to FIGS. 2C and 2D below.

[0076] The clamp **200** may include a movable upper jaw **210** and movable lower jaw **220** configured to interface with a variety of mounting structures (e.g. the mounting structure **122**) to couple the clamp **200** to the mounting structure such that, for example, components of the first and second manipulator assemblies **130** and **150** can be arranged and articulated with respect to the mounting structure. The upper jaw **210** may include an upper projection **212** to partially enclose a section of the mounting structure **122** in a lateral direction (e.g., the Y-direction) when the clamp **200** has a clamped mechanical state (is “clamped”) on the mounting structure **122**, as will be explained in greater detail below. As illustrated, the clamp **200** may include a pair of upper jaws **210** in a spaced apart configuration having an upper spacer **218** therebetween. In other embodiments, the clamp **200** has a continuous jaw portion and upper projection extending across the entire width of the upper jaw, or has any number of jaw segments and/or projections across the width of the upper jaw.

[0077] The upper jaw **210** may be movable with respect to an upper jaw housing **230** of the body **202**. In this regard, the upper jaw **210** may be slidably adjustable in at least one DOF with the upper jaw housing **230** by receiving an upper stem **232** coupled to the upper jaw **210**. As illustrated, a pair of upper stems **232** may be inserted through openings in the upper jaw housing **230** corresponding to each of the pair of upper jaws **210**. However, in other embodiments, any number of upper stems can be used to allow constrained movement of the upper jaws **210** with respect to the body **202**. The interface of the upper jaw **210** and upper stem **232** with the upper jaw housing **230** may include a biasing member **234** (e.g., a spring) configured to bias the upper jaw **210** and upper stem **232**, for example, away from the upper jaw housing **230**. The interface of the upper stem **232** and the upper jaw housing **230** may include a stop (e.g., a retaining clip, bushing, plug, etc. not shown) to prevent detachment of the upper stem **232** from the upper jaw housing **230** (e.g., as a result of the force of the biasing member **234** and/or other forces such as gravity). As will be explained in greater detail below, the biasing member **234** may be compressed as the upper jaw **210** translates with respect to the upper jaw housing **230** during attachment of the clamp **200** to a mounting structure.

[0078] The movable lower jaw **220** may include a lower projection **222** to partially enclose the

mounting structure **122** in a lateral direction when the clamp **200** is clamped on the mounting structure **122**. As illustrated, the clamp **200** may include a pair of lower jaws **220** in a spaced apart configuration having a lower spacer **228** therebetween. As with the upper jaw **210**, in other embodiments, the clamp **200** has a continuous jaw portion and lower projection extending across the entire width of the lower jaw, or has any number of jaw segments and/or projections across the width of the lower jaw. In some embodiments, the lower jaws **220** may be offset from the upper jaws **210** (e.g., in the X-direction).

[0079] The lower jaw **220** may be movable with respect to the body **202** by an articulable bracket **240** adjustable in at least one DOF with the body **202**. The lower jaw **220** may also be slidably adjustable in at least one DOF with the articulable bracket **240** by receiving a lower stem **242** (see FIGS. 2C and 2D) coupled to the lower jaw **220**. As illustrated, a pair of lower stems **242** may be inserted through openings in the articulable bracket **240** corresponding to each of the pair of lower jaws **220**. However, in other embodiments any number of lower stems can be used to allow constrained movement of the lower jaws **220** with respect to the articulable bracket **240**. The interface of the lower jaw **220** and lower stem **242** with the articulable bracket **240** may include a biasing member **244** (e.g., a spring) configured to bias the lower jaw **220** and lower stem **242**, for example, away from the articulable bracket **240**. The interface of the lower stem **242** and the articulable bracket **240** may include a stop (e.g., a retaining clip, bushing, plug, etc. not shown) to prevent detachment of the lower stem **242** from the articulable bracket **240** (e.g., as a result of the force of the biasing member **244** or other forces). As will be explained in greater detail below, the biasing member **244** may be compressed as the lower jaw **220** translates with respect to the articulable bracket **240** during attachment of the clamp **200** to a mounting structure.

[0080] FIGS. 2C-2F show cross-sectional right side views of the clamp **200**, the cross-sections taken along corresponding positions shown in FIGS. 2A and 2B. FIG. 2G shows a detail view of certain components of the clamp **200**, with the body **202** and the manual release lever **250** hidden for purposes of clarity. In some embodiments, the clamping articulation of the clamp **200** is accomplished by an electric motor **280** (see FIGS. 2E-2G) that moves the articulable bracket **240** to draw the jaws **210** and **220** toward the upper jaw housing **230** and the articulable bracket **240**, respectively, and inward toward the body **202** to engage the mounting structure **122** (shown in broken lines).

[0081] Referring to FIGS. 2C and 2D, the upper jaw **210** includes an upper lateral clamping surface **214** of the upper projection **212** and an upper vertical clamping surface **216**. The upper lateral clamping surface **214** and the upper vertical clamping surface **216** may be configured to abut corresponding surfaces of the mounting structure **122** during clamping of the clamp **200**. Similarly, the lower jaw **220** includes a lower lateral clamping surface **224** of the lower projection **222** and a lower vertical clamping surface **226**. The lower lateral clamping surface **224** and the lower vertical clamping surface **226** may be configured to abut corresponding surfaces of the mounting structure **122** during clamping of the clamp **200**. As shown in FIG. 2D, the upper and lower lateral clamping surfaces **214** and **224** may abut the same surface of the mounting structure **122**, e.g., a lateral or side surface, and the upper and lower vertical clamping surfaces **216** and **226** may abut opposite surfaces of the mounting structure **122**, e.g., the top and bottom surfaces. In this regard, the upper and lower lateral clamping surfaces **214** and **224** may capture the mounting structure **122** against a rail opposing surface **206** of the body **202** such that the mounting structure **122** has opposing pressure on both lateral sides when the clamp **200** is clamped.

[0082] Referring to FIGS. 2C-2G together, the articulable bracket **240** may be movable by the electric motor **280** with respect to the body **202**. The articulable bracket **240** may be configured to generally follow an arcuate path based on the pivoting movement of an upper cam **260** together with the lower cam **270**. As shown in FIGS. 2C and 2D, the upper cam **260** may be pivotably fixed at pivot A1 and the lower cam **270** may be pivotably fixed at pivot B1. In this regard, the upper and lower cams **260** and **270** may freely rotate about pivots A1 and B1, respectively, and such rotation

may be facilitated by an upper bearing **262** and a lower bearing **272**. In the illustrated embodiments, pivots **A1** and **B1** are fixed to at least the body **202** such that pivots **A1** and **B1** do not move with respect to the body **202** during clamping of the clamp **200**.

[0083] The upper cam **260** may include central pivot **A2** configured to pivotably interface with a translating rod **286** extending laterally from a junction **284** articulable by the electric motor **280** (see FIGS. 2E-2G). In some embodiments, the junction **284** is movable (e.g., in translation) with respect to the electric motor **280** by a rod **282**. The rod **282** may be threaded such that rotation of the electric motor **280** advances or retracts the junction **284** with respect to the electric motor **280**, thereby moving the translating rod **286** to effectuate rotation of the upper cam **260**. Central pivot **A2** may include an upper bearing **264** to facilitate rotation of the upper cam **260** with respect to the translating rod **286** during clamping of the clamp **200**. As shown in FIGS. 2C and 2D, the translating rod **286** may travel within an arcuate slot **265** during movement. Although the lower cam **270** is shown having an aperture for a central pivot point, the aperture may be included such that the same cam can be used in both the upper and lower positions to reduce part variation complexity, and the aperture in the lower cam **270** may not be used.

[0084] The upper cam **260** may be pivotably fixed to the articulable bracket **240** at pivot **A3**, and the lower cam **270** may be pivotably fixed to the articulable bracket **240** at pivot **B2**. In this regard, rotation of the upper and lower cams **260** and **270** about pivots **A3** and **B2**, respectively, may be facilitated by an upper bearing **266** and a lower bearing **274**. During movement of the upper and lower cams **260** and **270**, e.g., from the position shown in FIG. 2C to that shown in FIG. 2D, the position of the articulable bracket **240** may move while leaving the orientation of the articulable bracket **240** generally static with respect to the body **202**. In other embodiments, the upper and lower cams **260** and **270** may have differing lengths, pivots, and/or other geometry such that the orientation of the articulable bracket **240** changes with respect to the body **202** during movement. In these embodiments, certain of the components of the clamp **200**, e.g., the upper and lower cams **260** and **270**, the arcuate slot **265**, etc. may be positioned on either side of the clamp **200** in the X-direction (e.g. in pairs), such that a plurality of these components are used to articulate the clamp **200** during clamping to the mounting structure **122**. In other embodiments, components illustrated as having two or more instances, may be used with a single instance, or more instances than illustrated herein, and are within the scope of the present disclosure.

[0085] As described above, the manual release lever **250** may be used to override the clamped state (FIG. 2D) of the clamp **200** by pivoting the manual release lever **250** away from the body **202** in the direction of arrow **O**. As the manual release lever **250** is pivoted away from the body **202**, a locking pin **276** releases from a catch **277**, allowing the articulable bracket **240** to move from the position in FIG. 2D (clamped) to the position in FIG. 2C (an unclamped mechanical state, or “unclamped”).

[0086] With reference to FIGS. 2C-2F, movement of the components of the clamp **200** during one representative embodiment of clamping to a mounting structure **122** will now be described in greater detail. The upper jaw **210** and the lower jaw **220** have a translating DOF along axes of the upper stem **232** and the lower stem **242**, respectively. The DOF allows the upper and lower jaws **210** and **220** to linearly translate along the upper and lower stems **232** and **242** with respect to the upper jaw housing **230** and the articulable bracket **240**, respectively. Initially referring to the unclamped mechanical state shown in FIG. 2C, the upper and lower stems **232** and **242** are fully extended away from the upper jaw housing **230** and the articulable bracket **240**, respectively, as a result of the biasing force applied by the biasing members **234** and **244**. As the clamp **200** is placed on the rail, one or more sensors (not shown) on the clamp **200** may detect the rail and automatically initiate clamping of the clamp **200**. In other embodiments, a user, system, processor, etc. may initiate an automated or manual clamping operation of the clamp **200**.

[0087] As shown, the upper jaw **210** may be configured to travel along the axis of the upper stem **232** in a direction disposed at an angle α (see FIG. 2F) from vertical (e.g., along the opposing

surface **206**), and the lower jaw **220** may be configured to travel along the axis of the lower stem **242** in a direction disposed at an angle β from vertical. The angles α and β may be substantially the same angle from vertical, or the angles α and β may differ and be arranged at any suitable angle from vertical. In some embodiments, the angle α is between about 20° and 70° from vertical. In other embodiments, the angle α is between about 25° and 60° from vertical. In further embodiments, the angle α is between about 25° and 45° from vertical. Similarly, in some embodiments, the angle β is between about 20° and 70° from vertical. In other embodiments, the angle β is between about 25° and 60° from vertical. In further embodiments, the angle β is between about 25° and 45° from vertical.

[0088] As also shown in FIG. 2C, in the unclamped position, the articulable bracket **240** is positioned downward away from the mounting structure **122** such that the upper and lower cams **260** and **270** are rotated with pivots A3 and B2 in their lowest positions. In this configuration, the opening between the upper and lower jaws **210** and **220** may be maximized to allow ingress of the mounting structure **122** between the upper and lower jaws **210** and **220**. As a result of the arcuate path traveled by the pivots A3 and B2 during rotation of the upper and lower cams **260** and **270**, respectively, (e.g., along the arcuate slot **265**) the articulable bracket **240** may be positioned laterally closer to the body **202** than a position of the articulable bracket **240** at an intermediate point between the unclamped and clamped positions.

[0089] Referring to FIGS. 2C-2F, as the electric motor **280** moves the junction **284** upward in the Z-direction, the upper cam **260** rotates about pivot A1 to move the articulable bracket **240** along the arcuate slot **265**. For an initial portion of such rotation, the articulable bracket **240** moves upward, causing the lower jaw **220** to approach the mounting structure **122** in the Z-direction, and move laterally outward away from the opposing surface **206** in the negative Y-direction. At a point where a line drawn between pivot A1 and pivot A3 is horizontal, e.g., at the furthest lateral outboard position of the lower jaw **220**, further rotation will still cause the articulable bracket **240** to move upward, but will begin moving the lower jaw **220** laterally inward toward the opposing surface **206** via the rotating cams **260** and **270**.

[0090] When the clamp **200** is initially placed on the mounting structure **122**, the upper surface of the mounting structure **122** may contact the upper vertical clamping surface **216** of the upper jaw **210**. At a point along the arcuate path of the articulable bracket **240**, the lower vertical clamping surface **226** will contact the lower surface of the mounting structure **122**, and remain in contact during the remainder of the clamping movement. At this state, the upper and lower stems **232** and **242** will be extended from the upper jaw housing **230** and the articulable bracket **240**, respectively, such that the upper and lower lateral clamping surfaces **214** and **222** may be positioned away from the lateral surface of the mounting structure **122** (i.e., not contacting the lateral surface). As the articulable bracket **240** continues upward along the arcuate path, one or both of the upper and lower jaws **210** and **220** will begin to retract toward the upper jaw housing **230** and the articulable bracket **240**, respectively, to cause the jaws **210**, **220** to clamp to the mounting structure **122**.

[0091] In some embodiments, the biasing member **234** will have a force such that the upper jaw **210** retracts prior to the lower jaw **220**. The retraction of the upper jaw **210** toward the upper jaw housing **230** along the angle α will cause the upper lateral clamping surface **214** to approach the lateral outer surface of the mounting structure **122** (e.g., the surface facing away from the opposing surface **206** in the clamped mechanical state of the clamp **200**). As the upper jaw **210** is retracted, the clamp body **202** will move vertically downward with respect to the mounting structure **122** until the upper lateral clamping surface **214** abuts the lateral outer surface of the mounting structure **122**, laterally capturing the mounting structure **122** between the upper lateral clamping surface **214** and the opposing surface **206**. At this position, the upper jaw **210** will no longer retract toward the upper jaw housing **230**, and the vertical position of mounting structure **122** with respect to the clamp body **202** will not change.

[0092] After translation of the upper jaw **210** is complete and the upper lateral clamping surface

214 abuts the lateral outer surface of the mounting structure **122**, continued movement of the arcuate path of the articable bracket **240** will cause the lower jaw **220** to both travel toward the mounting structure **122** as a result of the arcuate path, and retract with respect to the articable bracket **240**. In a similar manner to the retraction of the upper jaw **210**, the arcuate path of the articable bracket **240** and the retraction of the lower jaw **220** toward the articable bracket **240** along the angle β will cause the lower lateral clamping surface **224** to approach the lateral outer surface of the mounting structure **122**, laterally capturing the mounting structure **122** between the lower lateral clamping surface **224** and the opposing surface **206**. At this clamped position, as shown in FIGS. 2D and 2F, the lower jaw **220** will no longer retract linearly toward the articable bracket **240**, movement of the articable bracket **240** by the electric motor **280** will stop, and the clamp **200** will be rigidly clamped to the mounting structure **122**. From the clamped position, the clamp **200** can be removed by reversing the electric motor **280** and/or pivoting the manual release lever **250**. Optionally, the clamp **200** may include a current sensor in electrical communication with the electric motor **280** to determine the clamp load that can be used to stop rotation of the electric motor **280**, detect a loosening of the rod **282**, etc. For example, the current sensor may detect an increase or spike in current to indicate that clamping has been successful and to stop continued actuation of the electric motor **280**.

[0093] FIGS. 3A and 3B are front-right-top and rear-right-top perspective views, respectively, of a modular clamp assembly (“clamp **300**”) configured in accordance with additional embodiments of the present disclosure. The clamp **300** is configured to be removably attached to a mounting structure such that various components and tools can be mounted to or supported by the clamp **300** and modularly arranged with respect to the mounting structure. In some embodiments, the mounting structure is a rail of the table T shown in FIGS. 1A and 1B, and the clamp **300** is configured to support one or more of the components described above in reference to the first and second manipulator assemblies **130** and **150** (FIG. 1B).

[0094] The clamp **300** generally includes a body **302** configured to at least partially surround and protect various components of the clamp **300**, provide component positioning and dynamic features, and include one or more accessory mounting locations, among other features. The body **302** may include a distal mounting location **304** configured to couple to other components of the system **100**, e.g. the coupling members **142** and **162**. In other embodiments, the body **302** includes any number of mounting locations to adapt the clamp **300** for mounting various other components. The body **302** may include other features, such as various slots, a cable receptacle **390** to allow routing of a cable within the body **302**, connector mounting features, indices **392** (e.g., indicator lamp, tactile indicators, etc.) to aid in positioning of the clamp **300**, etc. The clamp **300** may also include a manual release lever **350** (FIG. 3B) that can be pivoted in the direction of arrow O to release the clamp **300** from the mounting structure. The manual release lever **350** will be explained in greater detail with reference to FIGS. 3C and 3D below.

[0095] Routing of the cables through the cable receptacle **390** may allow the clamp **300** to act as a sterile adaptor, with a clean, sterile environment above the clamp **300**, and a non-sterile environment below the clamp **300**. The cable receptacle **390** may be configured to receive any suitable type of connector, including pass-through, cable-to-cable, pogo pin, intermediate PCB, etc. The cables within the cable receptacle **390** may be data, control, and/or power cables, and can be separated on either side of clamp. In some embodiments, the connectors and/or the cables may be recessed within the body **302** to avoid interfering with other components or staff. In embodiments where the clamp **300** acts as a sterile adapter, an upper end of the clamp **300** may be positioned at the height of the mounting structure **122** or higher, extending into the sterile area, and a lower end of the clamp **300** may be positioned below the mounting structure **122** in the non-sterile area (see, e.g., FIG. 5 of U.S. application Ser. No. 16/408,077, incorporated by reference herein). Although the cable receptacle **390** may not be shown with respect to the other embodiments described herein, the features are suitable for use with any of the embodiments of the present disclosure.

[0096] The clamp **300** may include a fixed upper jaw **310** and movable lower jaw **320** configured to interface with a variety of mounting structures (e.g., the mounting structure **122**) to couple the clamp **300** to the mounting structure such that, for example, components of the first and second manipulator assemblies **130** and **150** (FIG. **1B**) can be arranged and articulated with respect to the mounting structure. The upper jaw **310** may include an upper lateral clamping surface **312** (FIG. **3B**) configured to interface with a portion of the mounting structure **122** in a lateral direction (e.g., the Y-direction) when the clamp **300** is clamped on the mounting structure **122**, as will be explained in greater detail below. As illustrated, the clamp **300** may include a pair of upper jaws projections **314** (FIG. **3A**) in a spaced apart configuration. In other embodiments, the clamp **300** has a continuous upper jaw projection extending across the entire width of the upper jaw, or has any number of jaw segments and/or projections across the width of the upper jaw. The movable lower jaw **320** may project from an articulable bracket **340** to partially enclose the mounting structure **122** in a lateral direction when the clamp **300** is attached to the mounting structure **122**. As with the upper jaw **310**, in some embodiments, the clamp **300** may include any number of lower jaw segments and/or projections across the width of the lower jaw **320**. The lower jaw **320** may be movable with respect to the body **302** by the articulable bracket **340** associated in at least one DOF with the body **302**.

[0097] FIGS. **3C-3F** show cross-sectional right side views of the clamp **300**, the cross-sections taken along corresponding positions shown in FIGS. **3A** and **3B**. In some embodiments, the clamping articulation of the clamp **300** is accomplished by an electric motor **380** (see FIGS. **3E** and **3F**) that moves the articulable bracket **340** to draw the jaws **310** and **320** toward each other to clamp the mounting structure **122**. Referring initially to FIGS. **3C** and **3D**, the upper jaw **310** includes the upper lateral clamping surface **312** that may be configured to abut an upper corner of the mounting structure **122** during clamping of the clamp **300**. The upper lateral clamping surface **312** may be angled with respect to an opposing surface **306** of the body **302** such that the upper lateral clamping surface **312** can accommodate various widths of the mounting structure **122**. In this regard, rails with thinner widths would contact the upper lateral clamping surface **312** at a higher position along the upper jaw **310** than rails with thicker widths, which would contact the rail nearer a lower tip of the upper jaw **310**.

[0098] Similarly, the lower jaw **320** may include a lower lateral clamping surface **322**. The lower lateral clamping surface **322** may be configured to abut a lower corner of the mounting structure **122** during engagement of the clamp **300** to the rail. For example, as shown in FIG. **3D**, the upper and lower lateral clamping surfaces **314** and **322** may abut the upper and lower corners of the mounting structure **122**, respectively, e.g., the corners on the opposite side of the mounting structure **122** away from the opposing surface **306** in the clamped position. As described above with respect to clamp **200**, the jaws **210** and **220** of clamp **200** may contact the top, bottom, and two opposite lateral surfaces of the mounting structure **122**. In contrast with the clamp **200**, the clamp **300** may contact the mounting structure **122** on the upper and lower outer corners of the mounting structure **122** along with a lateral surface of the mounting structure **122** abutting the opposing surface **306**. In this regard, the upper and lower lateral clamping surfaces **314** and **322** may capture the mounting structure **122** against the rail opposing surface **306** such that the mounting structure **122** has opposing pressure on both lateral sides when the clamp **300** is attached to the mounting structure **122**.

[0099] Referring to FIGS. **3C-3F** together, the articulable bracket **340** may be movable upward in the Z-direction by the electric motor **380** with respect to the body **302**. The articulable bracket **340** may be configured to generally follow a segmented path based on the movement of an upper pin **342** within an upper slot **365** and a lower pin **344** within a lower slot **366**. As shown in FIGS. **3C** and **3D**, the upper slot **365** may have a segmented path with at least a first segment A and a second segment B arranged in different directions. The first and second segments A and B may be linear, may be any suitable shape (e.g., arcuate), may be combined into a single segment, or may be split

into more than two segments to create a suitable path of travel of the articulable bracket **340** during clamping of the clamp **300**.

[0100] The upper pin **342** may be configured to interface with a translating junction **384** articulable by the electric motor **380** (see FIGS. 3E and 3F). In some embodiments, the translating junction **384** is movable with respect to the electric motor **380** by a rod **382**. The rod **382** may be threaded such that rotation of the electric motor **380** advances or retracts the translating junction **384** with respect to the electric motor **380**, and thereby effectuate movement of the articulable bracket **340**. As shown in FIGS. 3C and 3D, the upper pin **342** may travel within the upper slot **365** and the lower pin **344** may travel within the lower slot **366** during movement of the articulable bracket **340**. Referring initially to the position of the articulable bracket **340** in FIG. 3C, the upper pin **342** will generally be positioned in the first segment A of the upper slot **365**. As shown, the first segment A is disposed at an angle from the second segment B. The orientation of the first segment A may be configured to move the lower jaw **320** substantially laterally with respect to the opposing surface **306**, and pivot about the lower pin **344** as a lower pin **344** travels within the lower slot **366**. In other embodiments, the orientation of the first segment A may be any suitable angle with respect to the second segment B.

[0101] As shown in the unclamped position of FIG. 3C, when the upper pin **342** is in the furthest position within the first segment A away from the opposing surface **306** and the lower pin **344** is in the lowest position within the lower slot **366**, the lower jaw **320** may be generally received within an indentation **308** within the body **302**. In this regard, it is possible to retract the articulable bracket **340** such that the lower jaw **320** does not protrude away from the body **302**, which may aid in attaching the clamp **300** to the mounting structure **122** prior to clamping (e.g., by providing additional clearance to mount the clamp **300** to rails of different sizes). As the articulable bracket **340** is moved away from the unclamped position in FIG. 3C to the clamped position in FIG. 3D, the upper pin **342** will travel along the first segment A to move the lower jaw **320** laterally outward with respect to the opposing surface **306** until the upper pin **342** reaches the second segment B. As the upper pin **342** travels along the second segment B, the upper pin **342** will generally travel in the same direction as the lower pin **344**. In the configuration shown, the articulable bracket **340** will generally initially rotate with respect to the lower pin **344** until the upper pin **342** reaches the second segment B, and then the articulable bracket **340** will generally translate linearly along the opposing surface **306**, as the upper pin **342** travels within the second segment B of the upper slot **365** and the lower pin **344** travels within the lower slot **366**. In these embodiments, certain of the components of the clamp **300**, e.g., the upper and lower slots **365** and **366**, the pins **342** and **344**, etc. may be positioned on either side of the clamp **300** in the X-direction (e.g. in pairs), such that a plurality of these components are used to articulate the clamp **300** during clamping to the mounting structure **122**. In other embodiments, components illustrated as having two or more instances, may be used with a single instance, or more instances than illustrated herein, and are within the scope of the present disclosure.

[0102] As mentioned above, the manual release lever **350** may be used to override the clamped state (FIG. 3C) of the clamp **300** by pivoting the manual release lever **350** about a pivot **358** away from the body **302** in the direction of arrow O. As the manual release lever **350** is pivoted away from the body **302**, a locking pin **352** releases from a catch **354**, allowing the articulable bracket **340** to move from the position in FIG. 3C (clamped) to the position in FIG. 3D (unclamped). When the locking pin **352** is released from the catch **354**, a manual release pin **356** may be configured to travel within a manual release slot **368**, moving the articulable bracket **340** downward and away from the mounting structure **122** to release the rail from the clamp **300**.

[0103] With reference to FIGS. 3C-3F, movement of the components of the clamp **300** during one representative embodiment of clamping to a mounting structure **122** will now be described in greater detail. Initially referring to the unclamped state shown in FIG. 3C, the clamp **300** will initially be placed on the mounting structure **122** by interfacing the upper jaw **310** with the

mounting structure **122**. In this configuration, the opposing surface **306** will contact the lateral surface of the mounting structure **122** facing the opposing surface **306**, and the upper lateral clamping surface **312** will contact an upper corner of the mounting structure **122**. As the clamp **300** is placed on the rail, one or more sensors (not shown) on the clamp **300** may detect the rail and automatically initiate clamping of the clamp **300**. In other embodiments, a user, system, processor, etc. may initiate an automated or manual clamping operation of the clamp **300**.

[0104] As also shown in FIGS. 3C and 3E, in the unclamped position, the articulable bracket **340** is positioned downward away from the mounting structure **122**. In this configuration, the opening between the upper and lower jaws **310** and **320** may be maximized, and the lower jaw **320** may be retracted within the indentation **308** to allow greater ingress of the mounting structure **122** between the upper and lower jaws **310** and **320**. As the electric motor **380** moves the translating junction **384** upward, the upper pin **342** travels along the first segment A toward the second segment B to substantially rotate the articulable bracket **340** about the lower pin **344**. During such movement along the first segment A, the lower pin **344** may have some component of movement within the lower slot **366**. However, in the illustrated embodiment, the movement of the lower pin **344** within the lower slot **366** will generally be smaller than the movement of the upper pin **342** within the first segment A, substantially resulting in a rotation of the articulable bracket **340** as opposed to a translation.

[0105] For an initial portion of the movement of the jaws **310** and **320**, the lower jaw **320** moves laterally outward away from the opposing surface **306** in the negative Y-direction. At a point where the upper pin **342** reaches the second segment B of the upper slot **365**, the rotation of the articulable bracket **340** will stop and further translation of the translating junction **384** will cause the articulable bracket **340** to move upward toward the mounting structure **122** in the Z-direction. As the articulable bracket **340** continues upward along the path of the second segment B and the lower slot **366**, the lower lateral clamping surface **322** will abut a lower corner of the mounting structure **122**, in a position shown in FIGS. 3C and 3E, where movement of the articulable bracket **340** by the electric motor **380** will stop, and the clamp **300** will be rigidly clamped to the mounting structure **122**. The clamp **300** may include a current sensor in electrical communication with the electric motor **380** to determine the clamp load that can be used to stop rotation of the electric motor **380**, detect a loosening of the rod **382**, etc. From the clamped position, the clamp **300** can be removed by reversing the electric motor **380** and/or pivoting the manual release lever **350**.

[0106] FIGS. 4A and 4B are front-right-top and rear-right-top perspective views, respectively, of a modular clamp assembly (“clamp **400**”) configured in accordance with additional embodiments of the present disclosure. In contrast with the clamps **200** and **300** described above with reference to FIGS. 2A-3F, the clamp **400** may be configured to clamp to the mounting structure **122** by manual articulation, as opposed to articulation by an electrical motor. The clamp **400** is configured to be removably attached to a structure such that various components and tools can be modularly arranged with respect to the structure. In some embodiments, the structure is the table T shown in FIGS. 1A and 1B, and the clamp **400** is configured to support one or more of the components described above in reference to the first and second manipulator assemblies **130** and **150** (FIG. 1B).

[0107] The clamp **400** generally includes a body **402** configured to at least partially surround and protect various components of the clamp **400**, provide component positioning and dynamic features, and include one or more accessory mounting locations, among other features. The body **402** may include a distal mounting location **404** configured to couple to other components of the system **100**, e.g., the coupling members **142** and **162**. In other embodiments, the body **402** includes any number of mounting locations to adapt the clamp **400** for mounting various other components. The body **402** may include other features, such as various slots, connector mounting features, indices (e.g., indicator lamp, tactile indicators, etc.) to aid in positioning of the clamp, etc. The clamp **400** may also include a lever **450** with a handle portion **451** (FIG. 4B) that can be pivoted in the direction of arrows CL and O to clamp and/or release the clamp **400**, respectively, from the

mounting structure, as will be explained in greater detail below.

[0108] As noted above with respect to clamp **300**, use of the clamp **400** may include routing of one or more cables through a cable receptacle (e.g., the cable receptacle **390** of the clamp **300** shown in FIGS. **3A** and **3B** may be suitable for use with the clamp **400**), and may allow the clamp **400** to act as a sterile adaptor, with a clean, sterile environment above the clamp **400**, and a non-sterile environment below the clamp **400**. Such cable receptacles may be configured to receive any suitable type of connector, including pass-through, cable-to-cable, pogo pin, intermediate PCB, etc. The cables within the cable receptacle may be data, control, and/or power cables, and can be separated on either side of clamp. In some embodiments, the connectors and/or the cables may be recessed within the body **402** to avoid interfering with other components or staff. In embodiments where the clamp **400** acts as a sterile adapter, an upper end of the clamp **400** may be positioned at the height of the mounting structure **122** or higher, extending into the sterile area, and a lower end of the clamp **400** may be positioned below the mounting structure **122** in the non-sterile area.

[0109] The clamp **400** may include a fixed upper jaw **410** and movable lower jaw **420** configured to interface with a variety of mounting structures (e.g., the mounting structure **122**) to couple the clamp **400** to the mounting structure such that, for example, components of the first and second manipulator assemblies **130** and **150** (FIG. **1B**) can be arranged and articulated with respect to the mounting structure. The upper jaw **410** may include an upper lateral clamping surface **412** (FIG. **4B**) configured to interface the mounting structure **122** in a lateral direction when the clamp **400** is clamped on the mounting structure **122**, as will be explained in greater detail below.

[0110] As illustrated, the clamp **400** may include a pair of upper jaws **410** (FIG. **4A**) in a spaced apart configuration. In other embodiments, the clamp **400** has a continuous upper jaw extending across the entire width of the upper jaw, or has any number of jaw segments and/or projections across the width of the upper jaw. The movable lower jaw **420** may project from an articulable bracket **440** movable with respect to a carriage **442** such that the lower jaw **420** is movable to partially enclose the mounting structure **122** in a lateral direction when the clamp **400** is clamped on the mounting structure **122**. As with the upper jaw **410**, in some embodiments, the clamp **400** may include any number of lower jaw segments and/or projections across the width of the lower jaw **420**. The lower jaw **420** may be movable with respect to the body **402** by the articulable bracket **440** and the carriage **442** associated in at least one DOF with the body **402**.

[0111] FIGS. **4C-4H** show cross-sectional right side views of the clamp **400**, the cross-sections taken along corresponding positions shown in FIGS. **4A** and **4B**. In some embodiments, the clamping articulation of the clamp **400** is accomplished by rotating the lever **450** (compare FIGS. **4C**, **4E**, and **4G** with **4D**, **4F**, and **4H**, respectively) that moves the carriage **442** to draw the jaws **410** and **420** together to clamp the mounting structure **122**. Referring initially to FIGS. **4C** and **4D**, the upper jaw **410** includes the first upper lateral clamping surface **412** and a second upper lateral clamping surface **414** that may be configured to abut a portion (e.g., a lateral surface, an upper corner, etc.) of the mounting structure **122** during clamping of the clamp **400**. The upper lateral clamping surfaces **412** and **414** may be angled or notched with respect to an opposing surface **406** of the body **402** such that the upper lateral clamping surfaces **412** and **414** can accommodate various widths of the mounting structure **122**. In this regard, rails with thinner widths can contact the first upper lateral clamping surface **412** and a higher position along the opposing surface **406** than rails with thicker widths, which would contact the second upper lateral clamping surface **414** nearer a tip of the upper jaw **410**.

[0112] Similarly, the lower jaw **420** may include a lower lateral clamping surface **422**. The lower lateral clamping surface **422** may be configured to abut a lower corner of the mounting structure **122** during clamping of the clamp **400**. As shown in FIG. **4D**, the upper and lower lateral clamping surfaces **412**, **414**, and **422** may abut one or more of the lateral surface or upper and lower corners of the mounting structure **122**, e.g., the corners positioned away from the opposing surface **406** in the clamped position. In contrast to the clamp **200**, the clamp **400** may contact the mounting

structure **122** on a lateral surface abutting the opposing surface **406** and the upper and lower outer corners and/or the lateral surface of the mounting structure **122** opposite the opposing surface **406**. In this regard, the upper and lower lateral clamping surfaces **412**, **414**, and **422** may capture the mounting structure **122** against the rail opposing surface **406** such that the mounting structure **122** has opposing pressure on both lateral sides when the clamp **400** is clamped.

[0113] Referring to FIGS. **4C-4H** together, the articuable bracket **440** and the carriage **442** may be movable by the lever **450** with respect to the body **402**. The articuable bracket **440** in the carriage **442** may be configured to move with respect to each other during clamping of the clamp **400**. In some embodiments, the movement of the articuable bracket **440** with respect to the carriage **442** may be biased by a biasing member **448** (see FIGS. **4A**, **4E**, and **4F**). As will be explained in greater detail below, the biasing member **448** may be configured to allow constrained movement of the articuable bracket **440** with respect to the carriage **442** to accommodate differing heights of the mounting structure **122**.

[0114] During rotation of the lever **450** in the direction of arrow O, the carriage **442** may be configured to generally follow a segmented arcuate path based on the movement of an upper pin **444** within an upper slot **465** and a lower pin **446** within a lower slot **466**. As shown in FIGS. **4C** and **4D**, the upper slot **465** may have a segmented path with at least a first segment A and a second segment B arranged in different directions, and the lower slot **466** may have a segmented path with at least a third segment C and a fourth segment D arranged in different directions. The first and second segments A and B may be substantially linear, may be any suitable shape (e.g., arcuate), may be combined into a single segment, or may be split into more than two segments to create a suitable path of travel of the carriage **442** during clamping of the clamp **400**. Likewise, the third and fourth segments C and D may be substantially linear, may be any suitable shape (e.g., arcuate), may be combined into a single segment, or may be split into more than two segments to create a suitable path of travel of the carriage **442** during clamping of the clamp **400**. In the illustrated embodiments, the upper and lower slots **465** and **466** have substantially the same length, orientation, and configuration. However, in other embodiments, the upper and lower slots **465** and **466** may have any suitable length, orientation, and configuration, and in these regards may differ between each other. As shown, the transition between the first and second segments A and B and between the third and fourth segments C and D may have an arcuate shape to smoothly transition the motion of the articuable bracket **440** and the carriage **442** as it transitions from travel within the second and fourth slots B and D to the first and third slots A and C during clamping of the clamp **400**.

[0115] As shown in FIGS. **4C** and **4D**, the upper pin **444** may travel within the upper slot **465** and the lower pin **446** may travel within the lower slot **466** during movement of the articuable bracket **440** and the carriage **442**. Referring initially to the position of the articuable bracket **440** and the carriage **442** in FIG. **4C**, the upper pin **444** will generally be positioned in the first segment A of the upper slot **465**, and the lower pin **446** will generally be positioned in the third segment C of the lower slot **466**. As shown, the first segment A is disposed at an angle from the second segment B and the third segment C is disposed at an angle from the fourth segment D. In other embodiments, the orientation of the first segment A may be any suitable angle with respect to the second segment B, and the orientation of the third segment C may be any suitable angle with respect to the fourth segment D.

[0116] The first and third segments A and C may control the travel of the articuable bracket **440** and the carriage **442** from the unclamped position (FIG. **4C**) as the clamp **400** articulates to the clamped position (FIG. **4D**). The orientations of the first and third segments A and C have a majority directional component in the horizontal Y-direction (as viewed in FIGS. **4C** and **4D**), with a minor directional component in the vertical Z-direction. Such orientations will cause the lower jaw **420** to translate substantially outward in the negative Y-direction, clearing the bottom surface of the mounting structure **122**, with a minor amount of translation in the Z-direction. As the upper

pin **444** and the lower pin **446** transition from the first and third segments A and C to the second and fourth segments B and D, the direction of travel of the articulable bracket **440** and the carriage **442** will change based on the orientation of the second and fourth segments B and D. In this regard, the orientations of the second and fourth segments B and D have a majority directional component in the vertical Z-direction, with a minor directional component in the horizontal Y-direction. Such orientations will cause the lower jaw **420** to translate substantially upward in the Z-direction, toward the bottom surface of the mounting structure **122**, with a minor amount of translation in the Y-direction such that the lower lateral clamping surface **422** translates toward the lateral surface of the mounting structure **122**.

[0117] As shown in the unclamped position of FIG. 4C, when the upper pin **444** is in the furthest position within the first segment A away from the opposing surface **406** and the lower pin **446** is also in the furthest position within the third segment C away from the opposing surface **406**, the lower jaw **420** may generally be positioned immediately adjacent to or abutting the body **402**. In this regard, it is possible to retract the articulable bracket **440** and the carriage **442** such that the lower jaw **420** minimally protrudes away from the body **402**, which may aid in attaching the clamp **400** to the mounting structure **122** prior to clamping. In other embodiments, the body **402** may have an indentation or recess (not shown) configured to receive the lower jaw **420** (see, e.g., the indentation **308** in FIG. 3C) such that the lower jaw **420** may be positioned to not protrude from the body **402**, to provide additional clearance for mounting the clamp **400** to mounting structures of different sizes. As described above, the lever **450** may be used to transition the clamp **400** between the clamped and unclamped positions by pivoting the lever **450** about a pivot **458** away from the body **402** in the direction of arrow O. As the lever **450** is pivoted away from the body **402**, a lever pin **476** begins to generally translate in the Y-direction, and eventually allows the articulable bracket **440** and the carriage **442** to move from the position in FIG. 4D (clamped) to the position in FIG. 4C (unclamped). The lever pin **476** may use an over-center locking design, as will be explained in greater detail below.

[0118] With reference to FIGS. 4C-4H, the movement of the components of the clamp **400** during one representative embodiment of clamping to a mounting structure **122** will now be described in greater detail. Initially referring to the unclamped state shown in FIGS. 4C, 4E, and 4G, the clamp **400** will initially be placed on the mounting structure **122** by interfacing the upper jaw **410** with the mounting structure **122**. In this configuration, the opposing surface **406** will contact the lateral surface of the mounting structure **122** facing the opposing surface **406**, and one or both of the upper lateral clamping surfaces **412** and **414** will contact at least an upper corner of the mounting structure **122**.

[0119] When the lever **450** is rotated in a direction of arrow CL as shown in FIG. 4C, the lever pin **476** is pivoted with the lever **450** about the pivot **458**, and may therefore follow an arcuate path away from the opposing surface **406**. The lever pin **476** may be operably coupled to a translating junction **484** through a telescoping piston assembly **452** configured to extend and retract such that the lever pin **476** and the translating junction **484** are movable with respect to each other during clamping of the clamp **400**. The translating junction **484** may be coupled to the upper pin **444** such that the translating junction **484** travels along the path of the upper slot **465** through the movement of the lever **450**. The telescoping piston assembly **452** may be dampened and/or biased (e.g., with a biasing member **432** surrounding the telescoping piston assembly **452**) such that movement of the lever pin **476** with respect to the translating junction **484** is restricted.

[0120] The damping and/or biasing of the telescoping piston assembly **452** may aid in clamping of the clamp **400** onto the mounting structure **122** and provide a force to hold the lever **450** against the body **402** in the clamped position using the over-center configuration shown in FIG. 4F. As shown, when the lever **450** is in the closed/clamped position, the lever pin **476** is positioned closer to the opposing surface **406** than a straight line extending through a central axis of the lever pivot **458** and a central axis of the upper pin **444** (e.g., the lever pin **476** is positioned on the side of the line

toward the negative Y-direction). In this configuration, the line extending through the lever pivot **458** and the upper pin **444** represents the direction of the force applied by the biasing member **432** tending to separate the translating junction **484** from the lever pin **476**. As a result of the position of the lever pin **476** in the over-center configuration, the lever **450** is biased toward the closed position until the lever **450** is rotated in the direction of arrow O such that the lever pin **476** is positioned further from the opposing surface **406** than a straight line connecting the lever pivot **458** and the upper pin **444** (e.g., the lever pin **476** crosses the line to the side of the line in the positive Y-direction). In other embodiments, the lever **450** may have a detent, a manual lock, or other suitable retention feature to keep the clamp **400** in the clamped position until removal.

[0121] In embodiments of the clamp **400**, certain of the components, e.g., the upper and lower slots **465** and **466**, the pins **444** and **446**, the telescoping piston assembly **452**, etc. may be positioned on either side of the clamp **400** in the X-direction (e.g., in pairs), such that a plurality of these components are used to articulate the clamp **400** during clamping to the mounting structure **122**. In other embodiments, components illustrated as having two or more instances, may be used with a single instance, or more instances than illustrated herein, and are within the scope of the present disclosure.

[0122] As also shown in FIGS. 4C, 4E, and 4G, in the unclamped position, the articulable bracket **440** and the carriage **442** are positioned downward away from the mounting structure **122** (see FIG. 4C). In this configuration, the opening between the upper and lower jaws **410** and **420** may be maximized, and the lower jaw **420** may be adjacent to the body **402** or optionally retracted within an indentation to allow ingress of the mounting structure **122** between the upper and lower jaws **410** and **420**. As the lever **450** rotates in a direction of arrow CL, the translating junction **484** and the upper pin **444** travel along the first segment A toward the second segment B and the lower pin **446** travels along the third segment C toward the fourth segment D to substantially translate the articulable bracket **440** and the carriage **442** in the direction of the first and third segments A and C.

[0123] For an initial portion of such movement, the lower jaw **420** translates in the negative Y-direction away from the body **402** with a minority component upward in the Z-direction toward the mounting structure **122**. At the transition point where the upper pin **444** reaches the second segment B of the upper slot **465** and the lower pin **446** reaches the fourth segment D of the lower slot **466**, the direction of translation of the articulable bracket **440** and the carriage **442** will transition to translation in the Z-direction toward the bottom surface of the mounting structure **122** with a minority component laterally inward in the Y-direction toward the lateral surface of the mounting structure **122**. As the articulable bracket **440** and the carriage **442** continues upward along the path of the second and fourth segments B and D. When the lower jaw **420** contacts the bottom surface of the mounting structure **122**, the lower lateral clamping surface **422** may not yet be in contact with the lateral surface of the mounting structure **122** four clamping of the clamp **400**. As described above, the movement of the articulable bracket **440** with respect to the carriage **442** may be biased by the biasing member **448**. As the lever **450** continues to be rotated to the closed position, the articulable bracket **440** will no longer move upward in the Z-direction and further upward movement of the carriage **442** will compress the biasing member **448**. This continued movement will only have a component in the Y-direction drawing the lower lateral clamping surface **422** toward the lateral surface of the mounting structure **122**.

[0124] After further rotation of the lever **450**, the lower lateral clamping surface **422** will abut the lateral surface of the mounting structure **122**, as shown in FIG. 4D. When the lower jaw **420** and the lower lateral clamping surface **422** are in contact with the mounting structure **122**, the lower jaw **420** will no longer move with respect to the mounting structure **122**, and the biasing members **432** and **448** will continue to compress as the lever **450** is rotated to the closed position abutting the body **402**. At this state, it may be desirable to lock the articulable bracket **440** and the carriage **442** with respect to each other, and the translating junction **484** and the lever pin **476** with respect to each other. Referring to FIGS. 4E and 4F, one embodiment of a toothed interface locking

mechanism for the translating junction **484** and the lever pin **476** is shown. A portion of the lever **450** may include teeth **434** protruding from a surface of the lever **450** and configured to contact a portion of the translating junction **484** when the lever **450** is in the closed position, as shown in FIG. **4F**. Correspondingly, the translating junction **484** may include teeth **436** configured to interface with the teeth **434** of the lever **450** to lock the translating junction **484** with respect to the lever pin **476**. By locking the components, the clamp **400** may be configured to withstand greater loads.

[0125] Referring now to FIGS. **4G** and **4H**, one embodiment of a locking mechanism for the articuable bracket **440** and the carriage **442** is shown. The upper pin **444** may include a rotatable arm **470** configured to interface with the lever **450** as the lever approaches the closed position. In this regard, the rotatable arm **470** may be biased away from the locked position, described below, until the lever **450** contacts the rotatable arm **470** and rotates the rotatable arm **470** toward the locked position. The rotatable arm **470** may include teeth **472** positioned at an end of the rotatable arm **470** and configured to contact teeth **474** positioned on the articuable bracket **440** to lock the articuable bracket **440** with respect to the carriage **442**. As the lever **450** rotates the rotatable arm **470**, the teeth **472** of the rotatable arm **470** engage with the teeth **474** of the articuable bracket **440** and lock the components. By locking the components, the clamp **400** may be configured to withstand greater loads. The clamp **400** may be removed from the mounting structure **122** by rotating the lever **450** in the direction of arrow O (see FIGS. **4D**, **4F**, and **4H**), in which the movement of the components of the clamp **400** described above will be substantially reversed.

[0126] FIGS. **5A** and **5B** are front-right-top and rear-right-top perspective views, respectively, of a modular clamp assembly (“clamp **500**”) configured in accordance with additional embodiments of the present disclosure. The clamp **500** is configured to be removably attached to a structure such that various components and tools can be modularly arranged with respect to the structure. In some embodiments, the structure is the table T shown in FIGS. **1A** and **1B**, and the clamp **500** is configured to support one or more of the components described above in reference to the first and second manipulator assemblies **130** and **150** (FIG. **1B**).

[0127] The clamp **500** generally includes a body **502** configured to at least partially surround and protect various components of the clamp **500**, provide component positioning and dynamic features, and include one or more accessory mounting locations, among other features. The body **502** may include a distal mounting location **504** configured to couple to other components of the system **100**, e.g. the coupling members **142** and **162**. In other embodiments, the body **502** includes any number of mounting locations, such as an intermediate mounting location **505** (FIG. **5B**), to adapt the clamp **500** for mounting various other components. The body **502** may include other features, such as various slots, such as a cable receptacle **590** to allow routing of a cable within the body **502**, connector mounting features for a connector **596**, such as a connector cavity **598**, indices (e.g., indicator lamp, tactile indicators, etc.) to aid in positioning of the clamp **500**, etc. The clamp **500** may also include a lever **550** with a handle portion **551** (FIG. **5B**) that can be pivoted in the direction of arrows CL and O to clamp and/or release the clamp **500**, respectively, from the mounting structure, as will be explained in greater detail below.

[0128] Routing of the cables through the cable receptacle **590** using connectors **596** may allow the clamp **500** to act as a sterile adaptor, with a clean, sterile environment above the clamp **500**, and a non-sterile environment below the clamp **500**. Such cable receptacles **590** may be configured to receive any suitable type of connector **596**, including pass-through, cable-to-cable, pogo pin, intermediate PCB, etc. The cables within the cable receptacle may be data, control, and/or power cables, and can be separated on either side of the clamp. In some embodiments, the connectors and/or the cables may be recessed within the body **502** to avoid interfering with other components or staff. In embodiments where the clamp **500** acts as a sterile adapter, an upper end of the clamp **500** may be positioned at the height of the mounting structure **122** or higher, extending into the sterile area, and a lower end of the clamp **500** may be positioned below the mounting structure **122**.

in the non-sterile area.

[0129] The clamp **500** may include a fixed upper jaw **510** and movable lower jaw **520** configured to interface with a variety of mounting structures (e.g., the mounting structure **122**) to couple the clamp **500** to the mounting structure such that, for example, components of the first and second manipulator assemblies **130** and **150** can be arranged and articulated with respect to the mounting structure. The upper jaw **510** may include an upper lateral clamping surface **512** (see FIG. 5C) configured to interface the mounting structure **122** in a lateral direction when the clamp **500** is clamped on the mounting structure **122**, as will be explained in greater detail below. As illustrated, the clamp **500** may include a continuous upper jaw **510** (FIG. 5A). In other embodiments, the clamp **500** has any number of jaw segments and/or projections across the width of the upper jaw. The movable lower jaw **520** may project from an articulable bracket **540** movable with respect to the body **502** such that the lower jaw **520** is movable to partially enclose the mounting structure **122** in a lateral direction when the clamp **500** is clamped on the mounting structure **122**. As with the upper jaw **510**, in some embodiments, the clamp **500** may include any number of lower jaw segments and/or projections across the width of the lower jaw **520**. The lower jaw **520** may be movable with respect to the body **502** by the articulable bracket **540** associated in at least one DOF with the body **502**.

[0130] FIGS. 5C-5H show cross-sectional right side views of the clamp **500**, the cross-sections taken along corresponding positions shown in FIGS. 5A and 5B. In some embodiments, the clamping articulation of the clamp **500** is accomplished by rotating the lever **550** (compare FIGS. 5C, 5E, and 5G with 5D, 5F, and 5H, respectively) that moves the articulable bracket **540** to draw the jaws **510** and **520** together to clamp the mounting structure **122**. Referring initially to FIGS. 5C and 5D, the upper jaw **510** includes the first upper lateral clamping surface **512** that may be configured to abut a portion (e.g., a lateral surface, an upper corner, etc.) of the mounting structure **122** during clamping of the clamp **500**. The upper lateral clamping surface **512** may be angled or notched with respect to an opposing surface **506** of the body **502** such that the upper lateral clamping surface **512** can accommodate various widths of the mounting structure **122**. In this regard, rails with thinner widths would contact the upper lateral clamping surface **512** at a higher position along the opposing surface **506** than rails with thicker widths, which would contact the rail nearer a tip of the upper jaw **510**.

[0131] Similarly, the lower jaw **520** may include a lower lateral clamping surface **522**. The lower lateral clamping surface **522** may be configured to abut a lower corner of the mounting structure **122** during clamping of the clamp **500**. As shown in FIG. 5D, the upper and lower lateral clamping surfaces **512** and **522** may abut one or more of the lateral surface or upper and lower corners of the mounting structure **122**, e.g., the corners positioned away from the opposing surface **506** in the clamped position. In contrast to the clamp **200**, the clamp **500** may contact the mounting structure **122** on a lateral surface abutting the opposing surface **506** and the upper and lower outer corners and/or the lateral surface of the mounting structure **122** opposite the opposing surface **506**. In this regard, the upper and lower lateral clamping surfaces **512** and **522** may capture the mounting structure **122** against the rail opposing surface **506** such that the mounting structure **122** has opposing pressure on both lateral sides when the clamp **500** is clamped.

[0132] Referring to FIGS. 5C-5H together, the articulable bracket **540** may be movable by the lever **550** with respect to the body **502**. During rotation of the lever **550** in the direction of arrow O, the articulable bracket **540** may be configured to generally follow a segmented arcuate path based on the movement of an upper pin **544** within an upper slot **565** and a lower pin **546** within a lower slot **566**. As shown in FIGS. 5C and 5D, the upper slot **565** may have a segmented path with at least a first segment A and a second segment B arranged in different directions, and the lower slot **566** may have a segmented path with at least a third segment C and a fourth segment D arranged in different directions. The first and second segments A and B may be substantially linear, may be any suitable shape (e.g., arcuate), may be combined into a single segment, or may be split into more

than two segments to create a suitable path of travel of the articulable bracket **540** during clamping of the clamp **500**. Likewise, the third and fourth segments C and D may be substantially linear, may be any suitable shape (e.g., arcuate), may be combined into a single segment, or may be split into more than two segments to create a suitable path of travel of the articulable bracket **540** during clamping of the clamp **500**. In the illustrated embodiments, the upper and lower slots **565** and **566** do not have the same length, orientation, or configuration. However, in other embodiments, the upper and lower slots **565** and **566** may have substantially the same length, orientation, and configuration, or any suitable length, orientation, and/or configuration. As shown, the transition between the first and second segments A and B and between the third and fourth segments C and D may have an arcuate shape to smoothly transition the motion of the articulable bracket **540** as the articulable bracket **540** transitions from travel within the second and fourth slots B and D to the first and third slots A and C during clamping of the clamp **500**.

[0133] As shown in FIGS. **5C** and **5D**, the upper pin **544** may travel within the upper slot **565** and the lower pin **546** may travel within the lower slot **566** during movement of the articulable bracket **540**. Referring initially to the position of the articulable bracket **540** in FIG. **5C**, the upper pin **544** will generally be positioned in the first segment A of the upper slot **565**, and the lower pin **546** will generally be positioned in the third segment C of the lower slot **566**. As shown, the first segment A is disposed at an angle from the second segment B and the third segment C is disposed at a different angle from the fourth segment D. In other embodiments, the orientation of the first segment A may be any suitable angle with respect to the second segment B, and the orientation of the third segment C may be any suitable angle with respect to the fourth segment D.

[0134] The first and third segments A and C may control the travel of the articulable bracket **540** from the unclamped position (FIG. **5C**) as the clamp **500** articulates to the clamped position (FIG. **5D**). The orientation of the first segment A has a directional component in the horizontal Y-direction and a directional component in the vertical Z-direction. Such an orientation will cause the lower jaw **520** to translate substantially outward in the negative Y-direction away from the opposing surface **506**, clearing the bottom surface of the mounting structure **122**, and travel upward in the Z-direction. The orientation of the third segment C has a directional component in the vertical Z-direction. Such an orientation will allow the articulable bracket **540** to translate substantially upward in the Z-direction as the lower jaw **520** rotates outward in the negative Y-direction.

[0135] As the upper pin **544** and the lower pin **546** transition from the first and third segments A and C to the second and fourth segments B and D, at the same or different times, the direction of travel of the articulable bracket **540** will change based on the orientation of the second and fourth segments B and D. In this regard, the orientation of the second segment B has a directional component in the horizontal Y-direction and a directional component in the vertical Z-direction. The orientation of the fourth segment D has a directional component in the horizontal Y-direction and a directional component in the vertical Z-direction. Such an orientation will cause the lower jaw **520** to continue to translate substantially upward in the Z-direction, toward the bottom surface of the mounting structure **122**, and the articulable bracket **540** to move in the Y-direction such that the lower lateral clamping surface **522** translates toward the lateral surface of the mounting structure **122**.

[0136] As shown in the unclamped position of FIG. **5C**, when the upper pin **544** is in the furthest position within the first segment A away from the opposing surface **506** and the lower pin **546** is also in the lowest position within the third segment C away from the opposing surface **506**, the lower jaw **520** may generally be positioned immediately adjacent to or abutting the body **502**. In this regard, it is possible to retract the articulable bracket **540** such that the lower jaw **520** minimally protrudes away from the body **502**, which may aid in attaching the clamp **500** to the mounting structure **122** prior to clamping. In other embodiments, the body **502** may have an indentation (not shown) configured to receive the lower jaw **520** (see e.g., the indentation **308** in

FIG. 3C) such that the lower jaw **520** may be positioned to not protrude from the body **502**. As described above, the lever **550** may be used to transition the clamp **500** between the clamped and unclamped positions by pivoting the lever **550** about a pivot **558** away from the body **502** in the direction of arrow O. As the lever **550** is pivoted away from the body **502**, a lever pin **554** begins to generally translate in the Y-direction, and eventually allows the articulable bracket **540** to move from the position in FIG. 5D (clamped) to the position in FIG. 5C (unclamped). The lever pin **554** may use an over-center locking design, as will be explained in greater detail below with respect to FIGS. 5G and 5H.

[0137] With reference to FIGS. 5C-5H, the movement of the components of the clamp **500** during one representative embodiment of clamping to a mounting structure **122** will now be described in greater detail. Initially referring to the unclamped state shown in FIG. 5C, the clamp **500** will initially be placed on the mounting structure **122** by interfacing the upper jaw **510** with the mounting structure **122**. In this configuration, the opposing surface **506** will contact the lateral surface of the mounting structure **122** facing the opposing surface **506**, and the upper lateral clamping surface **512** will contact an upper corner of the mounting structure **122**.

[0138] When the lever **550** is rotated in the direction of arrow CL (e.g., from FIG. 5C to 5D), the lever pin **554** is pivoted with the lever **550** about the pivot **558**, and may therefore cause the lower jaw **520** to follow an arcuate path away from the opposing surface **506**. The lever pin **554** may be operably coupled to a translating junction **584** through a telescoping piston assembly **552** configured to extend and retract such that the lever pin **554** and the translating junction **584** are movable with respect to each other during clamping of the clamp **500**. The translating junction **584** may be coupled to the upper pin **544** such that the translating junction **584** travels along the path of the upper slot **565** through the movement of the lever **550**. The telescoping piston assembly **552** may be dampened and/or biased (e.g., with a biasing member **532** (see FIGS. 5G and 5H) surrounding the telescoping piston assembly **552**) such that movement of the lever pin **554** with respect to the translating junction **584** is restricted.

[0139] The damping and/or biasing of the telescoping piston assembly **552** may aid in clamping of the clamp **500** onto the mounting structure **122** and provide a force to hold the lever **550** against the body **502** in the clamped position using the over-center configuration shown in FIG. 5H. As shown, when the lever **550** is in the closed/clamped position the lever pin **554** is positioned closer to the opposing surface **506** in the Y-direction than a straight line connecting the lever pivot **558** and the upper pin **544**. In this regard, the biasing member **532** exerts a force tending to separate the translating junction **585** from the lever pin **554**, and as a result of the position of the lever pin **554** in the over-center configuration, the lever **550** is biased toward the closed position until the lever **550** is rotated in the direction of arrow O such that the lever pin **554** is positioned further from the opposing surface **506** and the Y-direction than a straight line connecting the lever pivot **558** and the upper pin **544**. To aid in retention of the lever **550** in the closed position, as shown in FIG. 5G and 5H, the handle portion **551** may be configured to articulate about a pivot **580** and have a detent tooth **582** configured to interface with a detent tooth **588** (see FIG. 5E and 5F) of the body **502**. The articulation of the handle portion **551** may be biased to a locked position, engaging the detent teeth **582** and **588**, by a biasing member **586**. As the clamp **500** is transitioned from the clamped position to the unclamped position, the handle portion **551** would first be articulated about the pivot **580**, compressing the biasing member **586**, to release the detent teeth **582** and **588**, then the lever **550** would be rotatable with respect to the body **502**. In other embodiments, the lever **550** may have a manual lock or other suitable retention feature to keep the clamp **500** in the clamped position until removal.

[0140] In embodiments of the clamp **500**, certain of the components (e.g., the upper and lower slots **565** and **566**, the pins **544** and **546**, the telescoping piston assembly **552**, etc.) may be positioned on either side of the clamp **500** in the X-direction (e.g., in pairs), such that a plurality of these components are used to articulate the clamp **500** during clamping to the mounting structure **122**. In

other embodiments, components illustrated as having two or more instances, may be used with a single instance, or more instances than illustrated herein, and are within the scope of the present disclosure.

[0141] As also shown in FIG. 5C, in the unclamped position, the articulable bracket **540** is positioned downward away from the mounting structure **122**. In this configuration, the opening between the upper and lower jaws **510** and **520** may be maximized, and the lower jaw **520** may be adjacent to the body **502** or optionally retracted within an indentation to allow ingress of the mounting structure **122** between the upper and lower jaws **510** and **520**. As the lever **550** rotates in a direction of arrow CL, the translating junction **585** and the upper pin **544** travel along the first segment A toward the second segment B and the lower pin **546** travels along the third segment C toward the fourth segment D to substantially translate the articulable bracket **540** in the direction of the first and third segments A and C.

[0142] For an initial portion of such movement, the lower jaw **520** translates in the negative Y-direction away from the body **502** and upward in the Z-direction toward the mounting structure **122**. At the transition point where the upper pin **544** reaches the second segment B of the upper slot **565** and the lower pin **546** reaches the fourth segment D of the lower slot **566**, the direction of translation of the articulable bracket **540** will transition to translation in the Z-direction toward the bottom surface of the mounting structure **122** with a component laterally inward in the Y-direction toward the lateral surface of the mounting structure **122**. As the articulable bracket **540** continues upward along the path of the second and fourth segments B and D. When the lower jaw **520** contacts the bottom surface of the mounting structure **122**, the lower lateral clamping surface **522** may not yet be in contact with the lateral surface of the mounting structure **122** for clamping of the clamp **500**. As described above, the movement of the translating junction **584** with respect to the lever pin **554** may be biased by the biasing member **532**. As the lever **550** continues to be rotated to the closed position, the articulable bracket **540** will no longer move upward in the Z-direction and further upward movement of the lever pin **554** will compress the biasing member **532**. This continued movement will only have a component in the Y-direction drawing the lower lateral clamping surface **522** toward the lateral surface of the mounting structure **122**.

[0143] After further rotation of the lever **550**, the lower lateral clamping surface **522** will abut the lateral surface of the mounting structure **122**, as shown in FIG. 5D. When the lower jaw **520** and the lower lateral clamping surface **522** are in contact with the mounting structure **122**, the lower jaw **520** will no longer move with respect to the mounting structure **122**, and the biasing member **532** may continue to compress as the lever **550** is rotated to the closed position abutting the body **502** and the detent teeth **582** and **588** catch. The clamp **500** may be removed from the mounting structure **122** by rotating the lever **550** in the direction of arrow O (see FIGS. 5D, 5F, and 5H), in which the movement of the components of the clamp **500** described above will be substantially reversed.

[0144] The clamps **200**, **300**, **400**, and **500** described above may include various sensors (not shown) to transmit data signals regarding aspects of the clamp. Embodiments of the present disclosure may include one or more of the following: (1) strain gauges and/or load sensors to indicate an improper coupling and/or load on the clamp; (2) surface acoustic wave sensors (SAW) to determine residence related to security of the clamp on the rail; (3) an inductive coil for metal proximity; (4) a mechanical grounding sensor; (5) a mutual clamp proximity sensor (e.g., line of sight, laser rangefinder, LIDAR, etc.); (6) a rail orientation sensor, table angle sensor, and/or rail twist sensor (e.g., clamp-to-clamp, bending underload and/or oscillation, etc.); and (7) a user touch sensor to prevent an invalid state of the clamp. The clamps **200**, **300**, **400**, and **500** may also include various illumination features (not shown) to indicate certain aspects of the clamp, e.g., a locked state of the clamp, and unlock stated the clamp, overloading of the clamp, etc.

[0145] The above detailed descriptions of embodiments of the technology are not intended to be exhaustive or to limit the technology to the precise form disclosed above. Although specific

embodiments of, and examples for, the technology are described above for illustrative purposes, various equivalent modifications are possible within the scope of the technology, as those skilled in the relevant art will recognize. For example, while steps are presented in a given order, alternative embodiments may perform steps in a different order. Moreover, the various embodiments described herein may also be combined to provide further embodiments. Reference herein to “one embodiment,” “an embodiment,” or similar formulations means that a particular feature, structure, operation, or characteristic described in connection with the embodiment can be included in at least one embodiment of the present disclosure. Thus, the appearances of such phrases or formulations herein are not necessarily all referring to the same embodiment.

[0146] For case of reference, identical reference numbers are used to identify similar or analogous components or features throughout this disclosure, but the use of the same reference number does not imply that the features should be construed to be identical. Indeed, in many examples described herein, identically numbered features have a plurality of embodiments that are distinct in structure and/or function from each other. Furthermore, the same shading may be used to indicate materials in cross section that can be compositionally similar, but the use of the same shading does not imply that the materials should be construed to be identical unless specifically noted herein.

[0147] Moreover, unless the word “or” is expressly limited to mean only a single item exclusive from the other items in reference to a list of two or more items, then the use of “or” in such a list is to be interpreted as including (a) any single item in the list, (b) all of the items in the list, or (c) any combination of the items in the list. Where the context permits, singular or plural terms may also include the plural or singular term, respectively. Additionally, the term “comprising” is used throughout to mean including at least the recited feature(s) such that any greater number of the same feature and/or additional types of other features are not precluded. Directional terms, such as “upper,” “lower,” “front,” “back,” “vertical,” and “horizontal,” may be used herein to express and clarify the relationship between various elements. It should be understood that such terms do not denote absolute orientation. Further, while advantages associated with certain embodiments of the technology have been described in the context of those embodiments, other embodiments may also exhibit such advantages, and not all embodiments need necessarily exhibit such advantages to fall within the scope of the technology. Accordingly, the disclosure and associated technology can encompass other embodiments not expressly shown or described herein.

Claims

1. (canceled)
2. A clamping apparatus for mounting a surgical manipulator to a mounting structure, the clamping apparatus comprising: a body having a mounting location configured to support the surgical manipulator; an articulable bracket coupled to the body; a first jaw coupled to the body and configured to engage a first portion of the mounting structure; and a second jaw coupled to the articulable bracket and configured to engage a second portion of the mounting structure facing, wherein the articulable bracket is configured to articulate relative to the body along a path comprising multiple directions to move the second jaw from an unclamping position to a range of clamping positions, wherein the second jaw is spaced further from the body in the clamping positions than in the unclamping position, and wherein the second jaw is closer to the first jaw in the clamping positions than in the unclamping position.
3. The clamping apparatus of claim 2, wherein the path is an arcuate path.
4. The clamping apparatus of claim 2, wherein the path comprises multiple segments.
5. The clamping apparatus of claim 2, wherein: the first jaw further comprises one or more first clamping surface portions configured to abut a first portion of the mounting structure in the clamping position; and the second jaw further comprises one or more second clamping surface portions configured to abut a second portion of the mounting structure in the clamping position.

6. The clamping apparatus of claim 2, wherein one or both of the first jaw and the second jaw have an angled clamping surface portion to accommodate a range of widths of a mounting structure.
 7. The clamping apparatus of claim 2, wherein the first jaw is fixed relative to the body.
 8. The clamping apparatus of claim 2, wherein the first jaw is moveable relative to the body.
 9. The clamping apparatus of claim 2, wherein the second jaw is fixed relative to the articulable bracket.
 10. The clamping apparatus of claim 2, wherein the second jaw is moveable relative to the articulable bracket.
 11. The clamping apparatus of claim 2, wherein one or both of the first jaw and the second jaw are spring biased toward the other of the first jaw or the second jaw in the clamping positions.
 12. The clamping apparatus of claim 2, wherein the articulable bracket is motorized.
 13. The clamping apparatus of claim 13, further comprising a manually actuatable release lever operably coupled to the articulable bracket and configured to release the second jaw from a clamping position of the range of clamping positions.
 14. The clamping apparatus of claim 2, further comprising a manually actuatable lever operably coupled to the articulable bracket, wherein actuation of the lever causes articulation of the articulable bracket relative to the body.
 15. The clamping apparatus of claim 15, wherein the lever is moveable between a first position corresponding to the unclamping position of the second jaw, and a second position corresponding to a clamping position of the range of clamping positions, and wherein the lever has an over-center configuration in the second position.
 16. The clamping apparatus of claim 2, further comprising a sensor configured to detect a presence of a mounting structure.
 17. The clamping apparatus of claim 16, further comprising a motor configured to actuate the articulable bracket from the unclamping position to a clamping position of the range of clamping positions in response to detection of the presence of the mounting structure by the sensor.
 18. The clamping apparatus of claim 2, further comprising a sensor configured to detect a load on the clamping apparatus.
 19. The clamping apparatus of claim 2, wherein the body has a lateral surface extending between the first jaw and the second jaw and configured to be positioned opposing a third portion of the mounting structure extending between the first and second portions of the mounting structure.
 20. The clamping apparatus of claim 19, wherein the second jaw has a first range of motion away from the lateral surface of the body and a second range of motion toward the lateral surface of the body as the articulable bracket articulates relative to the body along the path to move the second jaw from the unclamping position to a clamping position of the range of clamping positions.
-